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PACKAGE STRUCTURE AND MANUFACTURING METHOD THEREOF

Abstract

A package structure includes a wafer-form semiconductor package and a thermal dissipating system. The wafer-form semiconductor package includes semiconductor dies electrically connected with each other. The thermal dissipating system is located on and thermally coupled to the wafer-form semiconductor package, where the thermal dissipating system has a hollow structure with a fluidic space, and the fluidic space includes a ceiling and a floor. The thermal dissipating system includes at least one inlet opening, at least one outlet opening and a plurality of first microstructures. The at least one inlet opening and the at least one outlet opening are spatially communicated with the fluidic space. The first microstructures are located on the floor, and at least one of the first microstructures is corresponding to the at least one outlet opening.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of and claims the priority benefit of a prior application Ser. No. 17/735,133, filed on May 3, 2022, now allowed. The prior U.S. application Ser. No. 17/735,133 is a continuation application of and claims the priority benefit of a prior application Ser. No. 16/921,916, filed on Jul. 6, 2020, now allowed, which claims the priority benefit of U.S. provisional application Ser. No. 62/907,692, filed on Sep. 29, 2019. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

[0002] As electronic products are continuously miniaturized, heat dissipation of the packaged semiconductor die(s) has become an important issue for packaging technology. In addition, for multi-die packages, the arrangement of the dies and the corresponding connecting elements has impact on data transmission speed among semiconductor dies and reliability of the packaged products.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 through FIG. 4 are schematic cross-sectional views illustrating intermediate structures formed at various stages of a manufacturing method of a (semiconductor) package structure in accordance with some embodiments of the disclosure.

[0005] FIG. 5 is a top view illustrating the package structure depicted in FIG. 4.

[0006] FIG. 6 through FIG. 13 are schematic cross-sectional views respectively illustrating a package structure in accordance with some embodiments of the disclosure.

[0007] FIG. 14 is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure.

[0008] FIG. 15 is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure.

[0009] FIG. 16 is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure.

[0010] FIG. 17A is a schematic cross-sectional view illustrating a package structure in accordance

with some embodiments of the disclosure.

[0011] FIG. 17B is an enlarged and schematic cross-sectional view illustrating a configuration of a portion of the heat dissipating module depicted in FIG. 17A.

[0012] FIG. 17C is an enlarged and schematic cross-sectional view illustrating a configuration of a portion of the heat dissipating module depicted in FIG. 17A.

[0013] FIG. 18A is a schematic top view illustrating the package structure depicted in FIG. 17A.

[0014] FIG. 18B and FIG. 18C are schematic top views respectively illustrating different parts of a heat dissipating module in the package structure depicted in FIG. 17A.

[0015] FIG. 19A is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure.

[0016] FIG. 19B is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure.

[0017] FIG. 20A is a schematic top view illustrating the package structure depicted in FIG. 19.

[0018] FIG. 20B and FIG. 20C are schematic top views respectively illustrating different parts of a heat dissipating module in the package structure depicted in FIG. 19.

[0019] FIG. 21 through FIG. 23 are schematic cross-sectional views illustrating intermediate structures formed at various stages of a manufacturing method of a package structure in accordance with some embodiments of the disclosure.

[0020] FIG. 24 through FIG. 26 are schematic cross-sectional views respectively illustrating a package structure in accordance with some embodiments of the disclosure.

[0021] FIG. 27 is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure.

[0022] FIG. 28A through FIG. 28B are schematic cross-sectional views respectively illustrating the configurations of microstructures of a heat dissipating system in a package structure in accordance with some embodiments of the disclosure.

[0023] FIG. 29A through FIG. 29B are schematic cross-sectional views respectively illustrating the configurations of microstructures of a heat dissipating system in a package structure in accordance with some embodiments of the disclosure.

[0024] FIG. 30 is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure.

[0025] FIG. 31A is a schematic three-dimensional side-view diagram of a heat dissipating module in a package structure in accordance with some embodiments of the disclosure.

[0026] FIG. 31B is a schematic top view illustrating the heat dissipating module depicted in FIG. 31A.

[0027] FIG. 31C is a schematic bottom view illustrating the heat dissipating module depicted in FIG. 31A.

[0028] FIG. 32 is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure.

[0029] FIG. 33 is a schematic cross-sectional view illustrating fluidic movements inside a heat dissipating system of a package structure in accordance with some embodiments of the disclosure.

[0030] FIG. 34 through FIG. 45 are schematic top views respectively illustrating an arrangement of microstructures and inflow/outflow channels of a heat dissipating system in a package structure in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

[0031] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components, values, operations, materials, arrangements, or the like, are described below to simplify the disclosure. These are, of course, merely examples and are not intended to be limiting. Other components, values, operations, materials, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows

may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. In the disclosure, it should be appreciated that the illustration of components throughout all figures is schematic and is not in scale.

[0032] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0033] In addition, terms, such as “first”, “second”, and the like, may be used herein for ease of description to describe similar or different element(s) or feature(s) as illustrated in the figures, and may be used interchangeably depending on the order of the presence or the contexts of the description.

[0034] Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

[0035] In some embodiments, for a semiconductor device such as a fan-out wafer level chip scale package (WLCSP), semiconductor dies are sawed from wafers before they are packaged onto other wafers, and only “known-good-dies” are packaged. An advantageous feature of this packaging technology is that the I/O pads on a semiconductor die can be redistributed to a greater area than the semiconductor die itself, and hence the number of I/O pads packed on the surfaces of the semiconductor dies can be increased. The disclosure will be described with respect to embodiments in a specific context, namely a package structure having a heat dissipating system which that realizes efficient heat dissipating for the fan-out WLCSP. The embodiments are not limited in this context. And, the embodiments are intended to provide further explanations but are not used to limit the scope of the disclosure. Sometimes, the fan-out WLCSP may also be referred to a semiconductor package of a wafer-form or a fan-out wafer-form semiconductor package.

[0036] In addition, the heat dissipating system, which realizes efficient heat dissipating for the fan-out WLCSP, includes one or more inlet channels and one or more outlet channels, where there is a microstructure (capable of slitting the flow of coolant) placed underneath the one or more outlet channels while there may also be or may not be another microstructure (capable of slitting the flow of coolant) placed underneath the one or more inlet channels. For illustrative purposes, the disclosure will be described with respect to embodiments with microstructures underneath both of the inlet and outlet channels. However, the disclosure is not specifically limited to have the microstructure placed underneath the one or more inlet channels of the heat dissipating system, but includes at least the microstructure placed underneath the one or more outlet channels of the heat dissipating system.

[0037] FIG. 1 through FIG. 4 are schematic cross-sectional views illustrating intermediate structures formed at various stages of a manufacturing method of a package structure in accordance with some embodiments of the disclosure. FIG. 5 is a top view illustrating the package structure

depicted in FIG. 4, where FIG. 1 through FIG. 4 are the cross-sectional views taken along a line AA' depicted in FIG. 5. FIG. 6 through FIG. 13 are schematic cross-sectional views respectively illustrating a package structure in accordance with some other embodiments of the disclosure. FIG. 33 is a schematic cross-sectional view illustrating fluidic movements inside a heat dissipating system of a package structure in accordance with some embodiments of the disclosure. Referring to FIG. 1, in some embodiments, a semiconductor package 200 is provided, and a metallization layer 300 is located over a backside surface S200b of the semiconductor package 200.

[0038] For example, the semiconductor package 200 includes a plurality of semiconductor dies 210, a plurality of input/output (I/O) interface dies 220, an insulating encapsulation 230, a redistribution circuit structure 240 and a plurality of conductive elements 250, as shown in FIG. 1. In some embodiments, the conductive elements 250 are the interfaces for external connections to the semiconductor package 200. That is, the conductive elements 250 serve as the conductive terminals of the semiconductor package 200 to electrical connect with the external devices/apparatus (e.g., a circuit carrier such as a circuit board (such as an organic circuit structure, a printed circuit board (PCB)), a system board, or the like; another semiconductor package or device; addition connectors; or the like) for electric signals output and/or input. In alternative embodiments, a semiconductor device (now shown) is optionally bonded to the semiconductor package 200 in a manner similar to the conductive elements 250. The semiconductor device may be an integrated passive element (IPD) or a surface mount device (SMD), the disclosure is not limited thereto.

[0039] In some embodiments, the semiconductor package 200 is a wafer-form (or wafer-scale or a wafer-size) semiconductor package. If considering a top view on a X-Y plane extending along a plane perpendicular to a stacking direction (e.g., direction Z) of the redistribution circuit structure 240 and the insulating encapsulation 230, the semiconductor package 200 may have a size (e.g., a diameter D shown in FIG. 5) about 4 inches or more. For example, the semiconductor package 200 has a diameter of about 6 inches. In some examples, the semiconductor package 200 has a diameter of about 8 inches. In some other examples, the semiconductor package 200 may have a diameter of about 12 inches.

[0040] The semiconductor dies 210 and the I/O interface dies 220 may be arranged aside to each other along the direction X. The semiconductor dies 210 and the I/O interface dies 220 may be arranged aside to each other along the direction Y. The direction X and the direction Y are not the same to each other and are not the same as the direction Z. For example, the direction X and the direction Y are perpendicular to each other and are respectively perpendicular to the direction Z. In some embodiments, the semiconductor dies 210 are arranged in the form of a matrix, such as a $N \times N$ array or a $N \times M$ array ($N, M > 0$, N may or may not be equal to M), while the I/O interface dies 220 are arranged to surround the semiconductor dies 210 (arranged into the array/matrix) for providing additional input/output circuitries thereto, and thus more I/O counts are provided to the semiconductor dies 210. The matrix of the I/O interface dies 220 may be a $N' \times N'$ array or a $N' \times M'$ array ($N', M' > 0$, N' may or may not be equal to M'). That is, in such embodiments, the I/O interface dies 220 are arranged into a matrix surrounding the perimeter of the matrix of the semiconductor dies 210.

[0041] However, the disclosure is not limited thereto, in an alternative embodiment, the semiconductor dies 210 and the I/O interface dies 220 are arranged in the form of a matrix, such as the $N'' \times N''$ array or $N'' \times M''$ arrays ($N'', M'' > 0$, N'' may or may not be equal to M''). With such embodiments, the semiconductor dies 210 and the I/O interface dies 220 are arranged into the matrix in an alternation manner. In a further alternative embodiment, the semiconductor dies 210 are arranged in the form of a first matrix and the I/O interface dies 220 are arranged in the form of a second matrix, where the first and second matrices are $N''' \times N'''$ array or $N'' \times M'''$ arrays ($N''', M''' > 0$, N''' may or may not be equal to M'''), and the first and second matrices are positioned next to each other along the direction X or the direction Y.

[0042] In some embodiments, the semiconductor dies **210** have a plurality of conductive vias **212**, where the conductive vias **212** serve as conductive terminals of the semiconductor dies **210** for electrical connection to other devices/elements (e.g., the redistribution circuit structure **240**). The semiconductor dies **210** each described herein may be referred to as a semiconductor chip or an integrated circuit (IC). For example, the semiconductor dies **210**, independently, are a logic chip, such as a central processing unit (CPU), graphics processing unit (GPU), system-on-chip (SoC), system-on-integrated-circuit (SoIC), microcontroller, or the like. However, the disclosure is not limited thereto; in alternative embodiments, the semiconductor dies **210**, independently, are a digital chip, analog chip or mixed signal chip, such as an application-specific integrated circuit (“ASIC”) chip, a sensor chips, a wireless and radio frequency (RF) chip, a baseband (BB) chip, a memory chip (such as high bandwidth memory (HBM) dies) or a voltage regulator chip. In further alternative embodiments, the semiconductor dies **210**, independently, are referred to as a chip or an IC of combination-type, such as a WiFi chip simultaneously including both of a RF chip and a digital chip. In some embodiments, a type of a first group of the semiconductor dies **210** are different from a type of a second group of the semiconductor dies **210**. In other words, the semiconductor dies **210** may include semiconductor chips or ICs of different types and/or the same type; the disclosure is not limited thereto.

[0043] In some embodiments, the I/O interface dies **220**, independently, have a plurality of conductive vias **222**, where the conductive vias **222** serve as conductive terminals of the I/O interface dies **220** for electrical connection to other devices/elements (e.g., the redistribution circuit structure **240**). As shown in FIG. 1, only four semiconductor dies **210** and two I/O interface dies **220** are presented for illustrative purposes, however, it should be noted that the number of the semiconductor dies **210** and the number of the I/O interface dies **220** may be one or more than one, the disclosure is not limited thereto.

[0044] In some embodiments, the semiconductor dies **210** and the I/O interface dies **220** are encapsulated in the insulating encapsulation **230**. The insulating encapsulation **230** laterally wrapped around the semiconductor dies **210** and the I/O interface dies **220**, where the conductive vias **212** of the semiconductor dies **210** and the conductive vias **222** of the I/O interface dies **220** are exposed by the insulating encapsulation **230**, for example, as shown in FIG. 1. In some embodiments, a sidewall of each conductive via **212** of the semiconductor dies **210** is partially covered (e.g. in physical contact with) by the insulating encapsulation **230**. In some embodiments, a sidewall of each conductive via **222** of the I/O interface dies **220** is partially covered (e.g. in physical contact with) by the insulating encapsulation **230**. For example, as shown in FIG. 1, illustrated top surfaces of the conductive vias **212** and the conductive vias **222** are substantially leveled with an illustrated top surface of the insulating encapsulation **230**. That is, in some embodiments, the illustrated top surfaces of the conductive vias **212**, the illustrated top surfaces of the conductive vias **222** and the illustrated top surface of the insulating encapsulation **230** are substantially coplanar to each other for achieving a high degree of coplanarity to facilitate the formation of a later-formed element (e.g., the redistribution circuit structure **240**).

[0045] However, the disclosure is not specifically limited. In alternative embodiments, the sidewall of each conductive via **212** of the semiconductor dies **210** is partially covered (e.g. in physical contact with) by the insulating encapsulation **230**, while the sidewall of each conductive via **222** of the I/O interface dies **220** is not covered by the insulating encapsulation **230**. In other embodiments, the sidewall of each conductive via **212** of the semiconductor dies **210** is not covered by the insulating encapsulation **230**, while the sidewall of each conductive via **222** of the I/O interface dies **220** is partially covered (e.g. in physical contact with) by the insulating encapsulation **230**. Or, in further alternative embodiments, the sidewall of each conductive via **212** of the semiconductor dies **210** is not covered by the insulating encapsulation **230**, while the sidewall of each conductive via **222** of the I/O interface dies **220** is also not covered by the insulating encapsulation **230**.

[0046] The insulating encapsulation **230** may include an acceptable insulating encapsulation material. The insulating encapsulation **230**, for example, includes polymers (such as epoxy resins, phenolic resins, silicon-containing resins, or other suitable resins), dielectric materials, or other suitable materials. The insulating encapsulation **230** may be a molding compound formed by a molding process. The insulating encapsulation **230** may further include inorganic filler or inorganic compound (e.g. silica, clay, and so on) which can be added therein to optimize coefficient of thermal expansion (CTE) of the insulating encapsulation **230**. The disclosure is not limited thereto.

[0047] In some embodiments, the redistribution circuit structure **240** is formed over the semiconductor dies **210**, the I/O interface dies **220** and the insulating encapsulation **230**. As shown in FIG. 1, the redistribution circuit structure **240**, for example, includes a fine-featured portion **240A** and a coarse-featured portion **240B**, and is electrically connected to the semiconductor dies **210** and the I/O interface dies **220** through connecting to the conductive vias **212** of the semiconductor dies **210** and the conductive vias **222** of the I/O interface dies **220** exposed by the insulating encapsulation **230**. In some embodiments, the fine-featured portion **240A** is located between the coarse-featured portion **240B** and the semiconductor dies **210** and between the coarse-featured portion **240B** and the I/O interface dies **220**. In some embodiments, the fine-featured portion **240A** of the redistribution circuit structure **240** is formed over and electrically coupled to the semiconductor dies **210** and the I/O interface dies **220**, and the coarse-featured portion **240B** is electrically coupled to the semiconductor dies **210** and the I/O interface dies **220** through the fine-featured portion **240A**. In some embodiments, as shown in FIG. 1, the fine-featured portion **240A** is capable of providing local electrical communications between the semiconductor dies **210**, between the I/O interface dies **220** and between the semiconductor dies **210** and the I/O interface dies **220**, while the coarse-featured portion **240B** is capable of providing global electrical communications between external devices/apparatus electrically connected to the conductive elements **250** and the semiconductor dies **210** and/or the I/O interface dies **220**.

[0048] For example, the fine-featured portion **240A** includes a dielectric structure **242A** and a metallization pattern **244A** located in the dielectric structure **242A**, and the coarse-featured portion **240B** includes a dielectric structure **242B** and a metallization pattern **244B** located in the dielectric structure **242B**. The metallization patterns **244A** and the metallization patterns **244B** independently may include one or more patterned conductive layers (which being individually referred to as redistribution layers or redistribution lines having line portions (also referred to as conductive lines or traces) on and extending on the X-Y plane and via portions (also referred to as conductive vias) extending on the direction Z), while the dielectric structures **242A** and the dielectric structures **242B** independently may include one or more dielectric layers arranged alternatively with the patterned conductive layers. The number of the dielectric layers included in one dielectric structure **242A** or **242B** and the number of the patterned conductive layers included in one metallization pattern **244A** or **244B** may not be limited to the drawings of the disclosure, and may be selected and designated based on the demand and design requirements.

[0049] The fine-featured portion **240A** and the coarse-featured portion **240B** of the redistribution circuit structure **240** include metallization patterns and dielectric structures of differing sizes, as shown in FIG. 1, for example. In certain embodiments, the patterned conductive layers included in the metallization pattern **244A** are formed from a same conductive material, and are formed to a same thickness (e.g., a first thickness) and a same line width (e.g., a first line width), and the patterned conductive layers included in the metallization pattern **244B** are formed from a same conductive material, and are formed to a same thickness (e.g., a second thickness) and a same line width (e.g., a second line width). Likewise, in some embodiments, the dielectric layers included in the dielectric structure **242A** are formed from a same dielectric material and are formed to a same thickness, and the dielectric layers included in the dielectric structure **242B** are formed from a same dielectric material and are formed to a same thickness. In some embodiments, along the direction Z, the patterned conductive layers included in the metallization pattern **244A** have the first

thickness that is smaller than the second thickness of the patterned conductive layers included in the metallization pattern **244B**. On the other hand, on the top view (e.g., on the X-Y plane), the patterned conductive layers included in the metallization pattern **244A** have the first line width that is smaller than the second line width of the patterned conductive layers included in the metallization pattern **244B**.

[0050] The material of the dielectric structures **242A**, **242B** may include polyimide, epoxy resin, acrylic resin, phenol resin, benzocyclobutene (BCB), polybenzoxazole (PBO), or any other suitable polymer-based dielectric material, and may be formed by deposition, lamination or spin-coating. The material of the metallization patterns **244A**, **244B** may include aluminum, titanium, copper, nickel, tungsten, and/or alloys thereof, and may be formed by electroplating or deposition. The disclosure is not limited thereto. The dielectric structures **242A**, **242B** and the metallization patterns **244A**, **244B** independently may also be patterned by a photolithography and etching process.

[0051] The material of the dielectric structure **242A** is, for example, as the same as the material of the dielectric structure **242B**. For another example, the materials of the dielectric structures **242A** and **242B** are different from one another. The material of the metallization pattern **244A** is, for example, as the same as the material of the metallization pattern **244B**. For another example, the materials of the metallization patterns **244A** and **244B** are different from one another. The disclosure is not limited thereto. In alternative embodiments, the redistribution circuit structure **240** may include metallization patterns of same size and dielectric structures of same size.

[0052] In some embodiments, the conductive elements **250** are attached to the redistribution circuit structure **240** for electrically coupling therebetween, as shown in FIG. **1**. For example, twelve conductive elements **250** are shown in FIG. **1** for illustrative proposes, however the number of the conductive elements **250** is not limited to the drawings of the disclosure, and may be selected and designed based on the demand. In some embodiments, through the redistribution circuit structure **240**, some of the conductive elements **250** are electrically connected to the semiconductor dies **210**, and some of the conductive elements **250** are electrically connected to the I/O interface dies **220**. Besides, some of the conductive elements **250** may be electrically floated or grounded, the disclosure is not limited thereto. The conductive elements **250** may be disposed on the redistribution circuit structure **240** by ball placement process or reflow process. The conductive elements **250** are, for example, solder balls or ball grid array (BGA) balls or bumps. Alternatively, the conductive elements **250** may include micro-bumps, metal pillars, electroless nickel-electroless palladium-immersion gold (ENEPIG) formed bumps, controlled collapse chip connection (C4) bumps, or the like; and may be formed by plating. The conductive elements **250** may be solder free. In the alternative embodiments of which the semiconductor device(s) is included, the semiconductor device may be disposed on the redistribution circuit structure **240** by flip-chip bonding technology or surface device mounting technology.

[0053] In some alternative embodiments (not shown), before disposing/forming the conductive elements **250** on the redistribution circuit structure **240**, a plurality of under-ball metallurgy (UBM) patterns are optionally formed on and electrically coupled to the redistribution circuit structure **240**, where the strength of connection between the conductive elements **250** and the redistribution circuit structure **240** is enhanced. The conductive elements **250** may be placed on the UBM patterns through ball placement process. That is, the conductive elements **250** may be electrically coupled to the redistribution circuit structure **240** through the UBM patterns. In some embodiments, the UBM patterns are made of a metal layer including a single layer or a metallization layer including a composite layer with a plurality of sub-layers formed of different materials. In some embodiments, the UBM patterns include copper, nickel, molybdenum, titanium, tungsten, titanium nitride, titanium tungsten, combinations thereof, or the like. For example, the UBM patterns include a titanium layer and a copper layer over the titanium layer. The UBM patterns may be formed using, for example, electroplating, sputtering, physical vapor deposition (PVD), or the like. However, the

UBM patterns may be omitted from the redistribution circuit structure **240**, the disclosure is not limited thereto.

[0054] Continued on FIG. **1**, in some embodiments, the metallization layer **300** is formed on the backside surfaces **S200b** of the semiconductor package **200**. In some embodiments, the metallization layer **300** may be a single layer or a composite layer having a plurality of sub-layers formed of different materials made of metal or metal alloy, which may include copper, nickel, titanium, tungsten, alloys thereof or other suitable choice of materials. The metallization layer **300** may be, for example, formed through a sputtering process, a PVD process, or the like. In some embodiments, the metallization layer **300** is conformally formed to be in direct contact with the backside surfaces **S200b** of the semiconductor package **200**. As shown in FIG. **1**, the metallization layer **300** is formed in a form of a blanket layer to be in physical contact with and covering illustrated backside surfaces **S210b** of the semiconductor dies **210**, illustrated backside surfaces **S220b** of the I/O interface dies **220** and an illustrated backside surface **S230b** of the insulating encapsulation **230**, where the metallization layer **300** is thermally coupled to and electrically isolated from the semiconductor dies **210** and the I/O interface dies **220**. In some embodiments, a thickness **T300** of the metallization layer **300** is approximately ranging from 5 μm to 1 mm.

[0055] Referring to FIG. **2**, in some embodiments, a plurality of bonding pads **310** and a bonding pad **402** are disposed on and distributed over the metallization layer **300**. For example, the bonding pads **310** and the bonding pad **402** are, disposed over the semiconductor package **200**, on a surface **S300** of the metallization layer **300**. The bonding pads **310** are surrounded by the bonding pad **402**, where the bonding pad **402** is in a closed frame shape, and the bonding pads **310** are placed within a region surrounded by the closed frame shape of the bonding pad **402**. In one embodiment, the bonding pads **310** are arranged in the form of a matrix, such as a $N'''\times N'''$ array or a $N'''\times M'''$ array (N''' , $M'''>0$, N''' may or may not be equal to M'''). In an alternative embodiment, the bonding pads **310** are arranged randomly. In a further alternative embodiment, the bonding pads **310** are arranged into a pre-determined pattern in correspondence of positioning locations of later-formed elements, e.g. microstructures (e.g. pillars **322**, splitters **324**, baffles **326**, fins **328** or combinations thereof, which will be further described later in FIG. **4**, FIG. **11**, FIG. **12**, and FIG. **13**). The shapes of the bonding pads **310** may be circular shapes, rectangular shapes, square shapes, ellipse shapes, polygonal shapes, or the like; the disclosure is not limited thereto.

[0056] A material of the bonding pads **310** may include a thermally conductive adhesive material, such as solder paste, for bonding with later-formed elements overlying thereto. In some embodiments, the bonding pads **310** is provided through a printing step (e.g., via stencil printing). For example, the bonding pads **310** include metallic pads made of solder paste by solder printing.

[0057] In some embodiments, the bonding pad **402** is formed in a manner of a continuous pad having an annular shape located on the metallization layer **300**. The closed frame shape of the bonding pad **402** may be in a circular, a rectangular, ellipse, or polygonal form. A material of the bonding pad **402** is not particularly limited, and may be chosen as a function of a material used for adhering the metallization layer **300** and a heat dissipating element (e.g. a heat dissipation module **500A** shown in FIG. **4**), where the bonding pad **402** has to secure the metallization layer **300** and the heat dissipation module **500A** together. For example, a material of the bonding pad **402** includes a thermo-curable adhesive, photocurable adhesive, thermally conductive adhesive, thermosetting resin, waterproof adhesive, lamination adhesive or a combination thereof. In some embodiments, the material of the bonding pad **402** includes a thermally conductive adhesive. For another example, the bonding pad **402** includes a die attach film (DAF). According to the type of material used, the bonding pad **402** may be formed by deposition, lamination, printing, plating, or any other suitable technique. In some embodiments, the material of the bonding pad **402** is different from the material of the bonding pads **310**. In some alternative embodiments, the material of the bonding pad **402** is the same as the material of the bonding pads **310**. In an alternative embodiment, the bonding pad **402** is an O-ring for wafer-edge sealing of the wafer-form

semiconductor package.

[0058] In some embodiments, the bonding pads **310** and the bonding pad **402** are formed in a same step. In some embodiments, the bonding pads **310** and the bonding pad **402** are formed in different steps. The bonding pads **310** may be formed prior to the formation of the bonding pad **402**. Alternatively, bonding pads **310** may be formed after the formation of the bonding pad **402**. The disclosure is not limited thereto.

[0059] Continued on FIG. 2, in some embodiments, after the formation of the bonding pads **310** and the bonding pad **402**, a microstructure **320**, such as a plurality of pillars **322**, is formed on the bonding pads **310** and are thermally coupled to the metallization layer **300** through the bonding pads **310**. For example, the pillars **322** are formed over (implanted on) the metallization layer **300** by performing pick-and-place pre-fabricated pillars onto the bonding pads **310** and performing soldering process to secure the pre-fabricated pillars and the bonding pads **310**. In some embodiments, a material of the pillars **322** includes metals or metal alloys. However, the disclosure is not limited thereto; in some alternative embodiments, the pillars **322** are directly grown on the metallization layer **300**, for example via a deposition or a plating step. For example, as shown in FIG. 2, the pillars **322** protrude directly from the metallization layer **300**. In the disclosure, the pillars **322** are referred to as the microstructure **320** for heat dissipating, as the pillars **322** are thermally coupled to the semiconductor package **200** via the bonding pads **310** and the metallization layer **300**.

[0060] In some embodiments, as show in FIG. 2, along the direction Z, thicknesses (e.g., T₃₂₂ depicted in FIG. 11) of the pillars **322** are the same. However, the disclosure is not limited thereto; alternatively, the thicknesses of the pillars **322** may be different from each other.

[0061] Referring to FIG. 3, in some embodiments, a heat dissipating module **500A** is provided and bonded to the metallization layer **300** through the bonding pad **402**. For example, the heat dissipating module **500A** includes a cover **510A** having a central portion **512A** and a flange portion **514A** at the periphery of the central portion **512A**. In some embodiments, the cover **510A** is bonded to the metallization layer **300** by connecting the flange portion **514A** to the boding pad **402**, where the bonding pad **402** is disposed on the metallization layer **300** only where the flange portion **514A** is expected to contact the metallization layer **300**. For example, the central portion **512A** is disposed over the semiconductor package **200** and extends substantially parallel to the metallization layer **300**. For example, the flange portion **514A** is disposed over the semiconductor package **200** and located at an edge of the central portion **512A**, and projected towards the metallization layer **300**. In some embodiments, the flange portion **514A** extends in a direction perpendicular to the plane defined by the central portion **512A**. For example, the flange portion **514A** is in contact with the edge of the central portion **512A**. In some embodiments, the flange portion **514A** and the central portion **512A** describe a right angle at their joint, but the disclosure is not limited thereto. In some embodiments, the flange portion **514A** is joined to the central portion **512A** at different angles than 90 degrees. In some embodiments, as illustrated in FIG. 3, the microstructure **320** (e.g., the pillars **322**) is surrounded by the flange portion **514A**, and are individually spacing apart from the central portion **512A** by a distance. However, the disclosure is not limited thereto, the microstructure **320** (e.g., the pillars **322**) may in contact with the central portion **512A** of the cover **510A**, alternatively.

[0062] In some embodiments, the metallization layer **300** and the heat dissipating module **500A** including the cover **510A** (having the central portion **512A** and the flange portion **514A**) together define an enclosure E1 of a heat dissipating system, where the enclosure E1 surrounds the microstructure **320** on all sides, and a fluidic channel **1001** is confined by the enclosure E1. In other words, the microstructure **320** is disposed inside the fluidic channel **1001**. In some embodiments, the enclosure E1 serves as an accommodating container of a coolant CL (shown in FIG. 4), where the coolant CL is in contact with the pillars **322** placed inside the fluidic channel **1001** to achieve heat transfer from the pillars **322** to the coolant CL for the purpose of heat dissipating.

[0063] The central portion **512A** may present regions of one or more thicknesses defining no recess, one recess or more recesses. For example, as shown in FIG. 4, when extending over the metallization layer **300**, the central portion **512A** presents one or more regions with a thickness **T512** (as measured along the direction Z) without recesses and one or more regions of another thickness **T1** (as measured along the direction Z) with recesses (e.g. **R1**). In some embodiments, a region of the thickness **T1** (which is smaller than the thickness **T512**) defines a circulation recess **R1** over the semiconductor package **200**. The thickness **T512** and the thickness **T1** may be selected and designated based on the demand and/or design requirement, the disclosure is not limited thereto; where the thickness **T1** is less than the thickness **T512**. In some embodiments, the circulation recesses **R1** individually are referred to as a fluidic storage or catchment. Owing to the recesses **R1**, more coolant **CL** is able to present inside the heat dissipating system for heat dissipation of the package structure **10A**. However, the disclosure is not limited thereto; in alternative embodiments, the central portion may have no recess or other recesses different from the recesses **R1** (e.g., FIG. 7 through FIG. 10, which will be further described later in detail).

[0064] The flange portion **514A** may present regions of one or more thicknesses defining no recess, one recess or more recesses. For example, as shown in FIG. 4, the flange portion **514A** presents a thickness **T514** (as measured along the direction Z) when extending towards the semiconductor package **200** without recesses. The thickness **T514** may be selected and designated based on the demand and/or design requirement, the disclosure is not limited thereto; where the thickness **T512** is less than the thickness **T514**. However, the disclosure is not limited thereto; the flange portion may also include one or more recesses (e.g., FIG. 6 and FIG. 7, which will be further described later in detail).

[0065] In some embodiments, the central portion **512A** includes one or more channels (e.g., **910** and **920**) spatially communicated with the fluidic channel **1001**. In the disclosure, the channel **910** may be referred to as one of an inflow channel and an outflow channel of the enclosure **E1** included in the heat dissipating system, and the channels **920** each may be referred to as other one of an inflow channel and an outflow channel of the enclosure **E1** included in the heat dissipating system, there is no specific limitations to the use of the channels **910** and **920**. Throughout the disclosure, for illustrative purposes (with the arrows **F** indicating the flow direction of coolant **CL** depicted in FIG. 4), the channel **910** is the outflow channel of the coolant **CL** while the channels **920** are the inflow channels of the coolant **CL**, for example. However, alternatively, the channel **910** may be the inflow channel of the coolant **CL** while the channels **920** may be the outflow channels of the coolant **CL**.

[0066] For illustrative purposes, only one channel **910** and only four channels **920** are shown; however, the number of the channel **910** and the number of the channel **920** may not be specifically limited to examples as described in the disclosure, and may be selected and designated based on the demand and/or design requirement. That is, the number of the inflow channels may be greater than, less than or substantially equal to the number of the outflow channels. For example, the number of the channel **910** is different from (e.g. less than (as shown in FIG. 3) or greater than (not shown)) the number of the channel **920**. For another example (not shown), the number of the channel **910** is the same as the number of the channel **920**.

[0067] In some embodiments, the channels (e.g. **910** and/or **920**) formed over the components of the semiconductor package **200** generating more heat during usage are used as outflow channels. However, the disclosure is not limited thereto; the channels (e.g. **910** and/or **920**) formed over the components of the semiconductor package **200** generating more heat during usage may be used as inflow channels. The positioning configurations of inflow/outflow channel may be determined based on the demand and design requirement (such as a hot spot inside the semiconductor package **200**), and may not be limited to the drawings as described in the disclosure.

[0068] In some embodiments, a size of the outflow channels may be at least as large as a size of the inflow channels. In alternative embodiments, the size of the outflow channels may be larger (has a

wider opening) than the size of the inflow channels. In further alternative embodiments, the size of the outflow channels may be smaller (has a narrow opening) than the size of the inflow channels. [0069] The inflow and/or outflow channels (e.g. **910**, **920**), one or more than one, may open with the circulation recess **R1**. For example, as shown in FIG. **3**, the channel **910** is not opening with the recess **R1** while the channels **920** are opening with the recesses **R1**. However, the disclosure is not limited thereto; the channels **910** and **920** may be, all or in part, not opening with the recess **R1** or with other types of recesses (such as a recess **R3** shown in FIG. **8**, a recess **R4** shown in FIG. **9** or the like).

[0070] For example, the channels **910**, **920** may be open holes having a substantially vertical profile in the thickness direction, but the disclosure is not limited thereto. In some embodiments, at least a portion of one of the channels **910**, **920** may run within the central portion **512A** along a direction tilted rather than vertical. In some embodiments, portions of the channels **910**, **920** may run parallel to the metallization layer **300**. As explained in further detail below, the channels **910**, **920** may be filled by the coolant **CL** (shown in FIG. **4**) flowing through the fluidic channel **1001** inside the enclosure **E1**.

[0071] In some embodiments, a material of the central region **512A** includes a thermally conductive material. In some embodiments, the material of the central region **512A** includes metals or metal alloys, such as copper, aluminum, their alloys, the combinations thereof or the like. In some alternatively embodiments, the material of the central region **512A** includes a plastic material, glass, ceramics, or the like. In some other embodiments, the material of the central region **512A** includes a semiconductor material such as silicon. In some embodiments, the material of the central region **512A** includes polyimide, epoxy resin, acrylic resin (e.g., polymethylmethacrylate, PMMA), phenol resin, BCB, PBO, or any other suitable polymer-based material. In some embodiments, a material of the flange portion **514A** may be selected from the same materials listed above for the central region **512A**.

[0072] In some embodiments, the material of the central region **512A** is the same as the material of the flange portion **514A**. For example, the central region **512A** and the flange portion **514A** are produced as a single piece (integrally formed). That is, the flange portion **514A** and the central region **512A** may be fabricated from the same material, and no interface or clear boundary may be visible between the flange portion **514A** and the central region **512A**. However, the disclosure is not limited thereto; the material of the flange portion **514A** may be different from the material of the central region **512A**. A method of forming the cover **510A** may be selected according to the material(s) chosen for the central region **512A** and the flange portion **514A**. In some embodiments, the cover **510A** is molded, forged, 3D-printed, grown, or fabricated according to any other suitable technique. In some embodiments, the central region **512A** and the flange portion **514A** are fabricated separately and then assembled to produce the cover **510A**.

[0073] Referring to FIG. **4**, in some embodiments, a fluid circulation system **700** is installed on the heat dissipating module **500A** in correspondence with the channels **910**, **920** to form a package structure **10A**. That is, for example, with the fluid circulation system **700**, the coolant **CL** flow (in and/or out of the enclosure **E1** of the heating dissipating system) is facilitated, thereby ensuring the heat dissipation of the package structure **10A**. The cover **510A** may be fabricated with washer(s) fitted into the channels **910**, **920** for subsequent connection with pipes or tubes. For example, the fluid circulation system **700** includes a plurality of pipes **710** connected to the cover **510A** of the heat dissipation module **500A** and, optionally, a plurality of washers **720**, that secure the attachment of the pipes **710** to the cover **510A**. In some embodiments, the pipes **710** and the channels **910**, **920** present matching treads (not shown), so that the pipes **710** can be securely fastened into the channels **910**, **920**, either directly or through intervening pipe connectors (not shown). The pipe connectors may include one-way valves that direct the flow of the coolant **CL** through the fluidic channel **1001**. That is, for example, the pipes **710** are connected with the channels **910**, **920** formed in the cover **510A** to allow the coolant **CL** to flow into the enclosure **E1** and to flow out of the

enclosure **E1** for removing heat generated by the semiconductor package **200** of the package structure **10A** during usage/operation.

[0074] In some embodiments, the coolant CL flows into the enclosure **E1** from the channels **920**, flows through the microstructure **320** inside the fluidic channel **1001**, and flows out of the enclosure **E1** from the channel **910**. In some embodiments, a stopper (not shown) may be used to seal any of the channels **910**, **920** if so required by the design of the fluid circulation system **700**. In some embodiments, the coolant CL is a liquid. In some embodiments, the coolant CL is water. Additives may be added to the water to produce a cooling fluid used as the coolant CL. Examples of additives include surfactants, corrosion inhibitors, biocides, antifreeze, and the like.

[0075] Continued on FIG. **4**, in some embodiments, an underfill **404** is formed on the metallization layer **300** exposed by the heat dissipating module **500A**. For example, the underfill **404** at least partially wraps sidewalls of the heat dissipating module **500A** (e.g., a sidewall of the flange portion **514A**). The underfill **404** may be any acceptable material, such as a polymer, epoxy, molding underfill, or the like, for example. In one embodiment, the underfill **404** is formed by underfill dispensing or any other suitable method. Owing to the underfill **404**, the bonding strength between the metallization layer **300** and the heat dissipating module **500A** is enhanced, thereby improving the reliability of the package structure **10A**.

[0076] Up to here, the package structure **10A**, which is installed with the heat dissipating system having the heating dissipating module **500A** and the metallization layer **300** with the microstructure **320** disposed thereon, is manufactured. In some embodiments, the microstructure **320** defines a network of interstices in fluidic communication inside the heat dissipating system. For example, some of the pillars **322** (e.g. the microstructure **320**) located on the metallization layer **300** are positioned underneath the openings of the channels **920**, where the coolant CL flows into the heat dissipating system through the channels **920** and is further divided into sub-flows of different directions due to the presence of such microstructure **320**. Owing to the microstructure **320** positioned underneath the openings of the channels in the heat dissipating system, the distribution of the coolant flow is improved, thereby enhancing the heat dissipation of the package structure **10A**.

[0077] However, the disclosure is not limited thereto. The heat dissipating system may be modified by adopting changes in the heat dissipating module and/or by adopting changes in the microstructures. The elements similar to or substantially the same as the elements described above will use the same reference numbers, and certain details or descriptions of the same elements (e.g. the formations and materials) and the relationship thereof (e.g. the relative positioning configuration and electrical connection) will not be repeated herein.

[0078] The non-limiting examples of the modifications to the heat dissipating module in the disclosure are shown in FIG. **6** to FIG. **10** for illustrative purposes, and are not intended to limit the scope of the disclosure. Referring to FIG. **4** and FIG. **6** together, the package structure **10A** depicted in FIG. **4** and a package structure **10B** depicted in FIG. **6** are similar; the difference is that, in the package structure **10B**, the heat dissipating module **500A** is substituted by a heat dissipating module **500B**.

[0079] As illustrated in the package structure **10B** of FIG. **6**, in some embodiments, the heat dissipating module **500B** includes a seal **520** in addition to a cover **510B** having a central portion **512B** and a flange portion **514B** at the periphery of the central portion **512B**. For example, the central portion **512B** and the flange portion **514B** are respectively similar to or the same as the central portion **512A** and the flange portion **514A** as described in FIG. **3**, and the difference is that the flange portion **514B** not only presents a region having the thickness **T514** (as measured along the direction **Z**) without recesses formed therein but also presents one or more regions of different thickness (e.g., **T2** as measured along the direction **Z**) with a recess (e.g., **R2**) formed therein. In some embodiments, a region of thickness **T2** smaller than the thickness **T514** defines an annular recess **R2** in the flange portion **514B** over the semiconductor package **200**, where the recess **R2** has

a shape similar to (in correspondence of) the shape of the flange portion **514B**. The thickness **T2** may be selected and designated based on the demand and/or design requirement, the disclosure is not limited thereto; where the thickness **T2** is less than the thickness **T514**. For example, thickness **T2** may be determined based on the thickness **T322** of the pillars **322** and/or a gap between the flange portion **514B** and a nearby pillar **322**. In some embodiments, the annular recess **R2** is referred to as a seal groove for accommodating the seal **520**, as shown in FIG. 6). In some embodiments, the seal **520** is accommodated within the annular recess **R2** and physically contacts (or is slightly compressed by) the central portion **512B** and the bonding pad **402** over the metallization layer **300**. In some embodiments, the seal **520** is a seal ring made of a polymeric material, such as an organic resin or rubber, and provides closure and segregation for avoiding fluid leakage from the enclosure **E1**. In some embodiments, the seal **520** may include a silicone filling. [0080] Alternatively, the bonding pad **402** may be optionally omitted from the package structure **10B**. In such alternative embodiments, a clamping fixture (not shown) is used to providing external pressure to secure the metallization layer **300** and the heat dissipation module **500B** together. The clamping fixture may have a configuration similar to a supporting module **600** (will be described later in conjunction with FIG. 16).

[0081] The recesses **R1** may be omitted from the package structure **10B** (see a package structure **10C** depicted in FIG. 7) or from the package structure **10A** (see a package structure **10F** depicted in FIG. 10). As illustrated in the package structure **10C** of FIG. 7, in some embodiments, a heat dissipating module **500C** includes a cover **510C** having a central portion **512C** and a flange portion **514C** at the periphery of the central portion **512C**. For example, the central portion **512C** and the flange portion **514C** are respectively similar to or the same as the central portion **512B** and the flange portion **514B** as described in FIG. 6, and the difference is that the central portion **512C** has no recess (e.g. **R1**). As illustrated in the package structure **10F** of FIG. 10, in some embodiments, a heat dissipating module **500F** includes a cover **510F** having a central portion **512F** and a flange portion **514F** at the periphery of the central portion **512F**. For example, the central portion **512F** and the flange portion **514F** are respectively similar to or the same as of the central portion **512A** and the flange portion **514A** as described in FIG. 4, and the difference is that the central portion **512F** has no recess (e.g. **R1**).

[0082] Alternatively, additional recesses may be included to the package structure **10A** (see the recess **R3** shown in a package structure **10D** depicted in FIG. 8 and a recess **R4** shown in a package structure **10E** depicted in FIG. 9). As illustrated in the package structure **10D** of FIG. 8, for example, a heat dissipating module **500D** includes a cover **510D** having a central portion **512D** and a flange portion **514D** at the periphery of the central portion **512D**. For example, the central portion **512D** and the flange portion **514D** are respectively similar to or the same as the central portion **512A** and the flange portion **514A** as described in FIG. 4, and the difference is that the central portion **512D** further include at least one recess **R3** connected to the channel **910**. That is, for example, as shown in FIG. 8, the central portion **512D** not only presents the region of the thickness **T512** without recesses and the regions of the thickness **T1** with the recesses **R1**, but also presents one or more regions of a thickness **T3** extending over the metallization layer **300** with one or more recesses (e.g., **R3**) formed therein. In some embodiments, a region of thickness **T3** smaller than the thickness **T512** defines a circulation recess **R3** over the semiconductor package **200**. That is, for example, as shown in FIG. 8, the channel **910** opens with and spatially communicated with the recess **R3**. In some embodiments, the circulation recess **R3** is referred to as a fluidic storage or catchment. Owing to the recess **R3**, more coolant **CL** is able to present inside the heat dissipating system for heat dissipation of the package structure **10D**. The thickness **T3** may be selected and designated based on the demand and/or design requirement, the disclosure is not limited thereto; where the thickness **T3** is less than the thickness **T512**. However, the disclosure is not limited thereto; in alternative embodiments, the central portion **512D** may have no other recess except the recess **R3**.

[0083] In other embodiments, as illustrated in the package structure **10E** of FIG. **9**, a heat dissipating module **500E** includes a cover **510E** having a central portion **512E** and a flange portion **514E** at the periphery of the central portion **512E**. For example, the central portion **512E** and the flange portion **514E** are respectively similar to or the same as the central portion **512A** and the flange portion **514A** as described in FIG. **4**, and the difference is that the central portion **512E** further include at least one recess **R4**. That is, for example, as shown in FIG. **9**, the central portion **512E** not only presents the region of the thickness **T512** without recesses and the regions of the thickness **T1**, but also presents one or more regions extending over the metallization layer **300** with one or more recesses (e.g., **R4**) formed therein. The thickness **T4** may be selected and designated based on the demand and/or design requirement, the disclosure is not limited thereto; where the thickness **T4** is less than the thickness **T512**. The recesses **R4** may be additionally included in the central portion **512E** or adopted for replacing at least one recess **R1** depicted in FIG. **4**. For example, as shown in FIG. **9**, the channel **920** opens with and spatially communicated with the recess **R4** instead of the recess **R1**, while additional recesses **R4** are further adopted to ensure the coolant flow distribution and movement. In some embodiments, the recesses **R4** are individually referred to as a fluidic slot. Owing to the recesses **R4**, not only more coolant **CL** is able to present inside the heat dissipating system for heat dissipation of the package structure **10D**, but also the coolant flow distribution and movement is further ensured. However, the disclosure is not limited thereto; in alternative embodiments, the central portion **512E** may have no other recess except the recesses **R4**.

[0084] The non-limiting examples of the modifications to the microstructures in the disclosure are shown in FIG. **11** to FIG. **13** for illustrative purposes, and are not intended to limit the scope of the disclosure. For example, a package structure **10G** depicted in FIG. **11** is similar to the package structure **10A** depicted in FIG. **4**, and the difference is that, instead of the microstructure **320**, the package structure **10G** includes a microstructure **320A**. In some embodiments, the microstructure **320A** includes a plurality of pillars **322** and a plurality of splitters **324**, where the splitters **324** are at least positioned underneath the openings of the channels **910**, **920**. Due to the splitters **324**, the distribution of the coolant flow is further improved, thereby further enhancing the heat dissipation of the package structure **10G**. In some embodiments, the pillars **322** individually has a constant width **W322** as measured along the direction **X** and a thickness **T322** as measured along the direction **Z**, where the width **W322** and the thickness **T322** may be selected and designated based on the demand and/or design requirement, the disclosure is not limited thereto. For example, the pillars **322** are micro-pillars having the constant width along the direction **Z** from the metallization layer **300** toward the central portion **512A**. In some embodiments, the splitters **324** individually has a non-constant width **W324** as measured along the direction **X** and a thickness **T324** as measured along the direction **Z**, where the width **W324** and the thickness **T324** may be selected and designated based on the demand and/or design requirement, the disclosure is not limited thereto. For example, the splitters **324** are micro-pillars having the width increased along the direction **Z** from the metallization layer **300** toward the central portion **512A**. The material and formation of the pillars **322** have been described in FIG. **2**, and thus are not repeated herein for brevity. The formation and material of the splitters **324** may be the same or similar to the formation and material of the pillars **322**. Similar to the pillars **322**, the splitters **324** are individually spacing apart from the central portion **512A** by a distance, for example.

[0085] Alternatively, the microstructure **320** may be substituted by a microstructure **320B**. For example, a package structure **10H** depicted in FIG. **12** is similar to the package structure **10A** depicted in FIG. **4**, and the difference is that, instead of the microstructure **320**, the package structure **10H** includes the microstructure **320B**. In some embodiments, the microstructure **320B** includes a plurality of pillars **322** and a plurality of baffles **326**, where the baffles **326** are free from portions of the metallization layer **300** underneath the openings of the channels **910**, **920**. In some embodiments, two adjacent baffles **326** together confine a partition space (not labeled) inside the

fluidic channel **1001**. That is, two adjacent baffles **326** define a micro-chamber in between. The number of the partition spaces inside the fluidic channel **1001** is not limited in the disclosure, and may be designated and selected based on the demand. Due to the baffles **326**, the movement of the coolant flow is controllable, which leads to a fast heat dissipation, thereby further enhancing the heat dissipation of the package structure **10H**. As shown in FIG. **12**, the baffles **326** individually are bonded to the metallization layer **300** through the bonding pads **310** and bonded to the central portion **512A** through bonding pads **530**, for example. In some embodiments, the baffles **326** individually has a constant width (not labeled) as measured along the direction X, where the width of the baffles **326** may be selected and designated based on the demand and/or design requirement, the disclosure is not limited thereto. The formation and material of the baffles **326** may be the same or similar to the formation and material of the pillars **322**, and thus are not repeated herein. In some further embodiments, the microstructure **320B** further includes the splitters **324**.

[0086] Furthermore, the microstructure **320** may be substituted by a microstructure **320C**. For example, a package structure **10I** depicted in FIG. **13** is similar to the package structure **10A** depicted in FIG. **4**, and the difference is that, instead of the microstructure **320**, the package structure **10I** includes the microstructure **320C**. In some embodiments, the microstructure **320C** includes a plurality of pillars **322** and a plurality of fins **328**, where the fins **328** are free from portions of the metallization layer **300** underneath the openings of the channels **910**, **920**. In some embodiments, two adjacent fins **328** together confine a serpentine path (not labeled) inside the fluidic channel **1001**. That is, two adjacent fins **328** define a micro-trench in between. The number of the serpentine path inside the fluidic channel **1001** is not limited in the disclosure, and may be designated and selected based on the demand. Due to the fins **328**, the movement of the coolant flow is further controllable, thereby further enhancing the heat dissipation of the package structure **10I**. As shown in FIG. **13**, the fins **328** individually are bonded to the metallization layer **300** through the bonding pads **310** and spacing apart from the central portion **512A** via a distance to allowing the coolant CL passing therethrough, for example. In some embodiments, the fins **328** individually has a constant width (not labeled) as measured along the direction X, where the width of the baffles **326** may be selected and designated based on the demand and/or design requirement, the disclosure is not limited thereto. The formation and material of the fins **328** may be the same or similar to the formation and material of the pillars **322**, and thus are not repeated herein. In some further embodiments, the microstructure **320C** further includes the splitters **324**, the baffles **326**, or a combination thereof.

[0087] For example, the flow movements of the coolant CL due to the presences of the microstructures **320**, **320A**, **320B**, or **320C** (e.g., **322**, **324**, **326**, **328**) inside the heat dissipating system are demonstrated in FIG. **33** for illustrative purposes, where an arrangement of the microstructures (e.g., **322**, **324**, **326**, **328**) is not specifically limited thereto. The non-limiting examples of possible arrangements of the microstructures (e.g., **322**, **324**, **326**, **328**) will be discussed in view of FIG. **34** through FIG. **45** (further discussed in detail below). Referring to FIG. **33**, in some embodiments, if assuming the coolant CL proceeds from the right side of the drawing towards the left side of the drawing (along the direction X), the incident flow (indicated by arrows) is deviated by the pillars **322** and the splitters **324**, forced to pass above the fins **328**, and deviated by the baffle **326**. In some embodiments, the incident flow of the coolant CL flows in a space between two baffles **326**, between two fins **328** or between one baffle **326** and one fin **328**, and passes above the pillars **322** and the splitters **324**. As show in FIG. **33**, the microstructures (e.g., **322**, **324**, **326**, **328**) are separated from each other on the X-Y plane, for example. In other words, the microstructures (e.g., **322**, **324**, **326**, **328**) are not overlapped to each other in direction Z. In addition, the presences of the microstructures (e.g., **322**, **324**, **326**, **328**) along with the metallization layer **300** thermally coupled to the semiconductor package **200** greatly increases the heat dissipation in the direction Z for the semiconductor package **200**.

[0088] It is appreciated that in the disclosure, the recesses **R1**, **R2**, **R3**, **R4** or combinations thereof

along with the microstructures (e.g., **322**, **324**, **326** and **328**) may be adopted to the heat dissipating system based on the demand and design layout, as long as there is at least one micro-pillar (e.g., **322** or **324**) disposed at a location corresponding to (e.g., underneath) the opening of each of the inflow/outflow channels of the heat dissipating system for splitting the coolant CL into different sub-flows. The numbers of the recesses **R1**, **R2**, **R3** and **R4** and the numbers of the microstructures (e.g., **322**, **324**, **326** and **328**) are not limited to the drawings of the disclosure, and may be selected and designated based on the demand and design layout. In some embodiments, the microstructures (e.g., **322**, **324**, **326** and **328**) may be coated with a thermally conductive material. In some embodiments, the coating material includes graphene or diamond-like carbon (DLC). In some alternative embodiments, the coating material includes metals such as copper or aluminum.

[0089] The package structures **10A** through **10I** may be further mounted with a circuit component (such as a circuit substrate) or other electronic devices to form a stacked package structure through the conductive elements **250** and/or other additional connectors based on the design layout and the demand. For illustration, a non-limiting example is provided as a package structure **10J** of a stacked structure in FIG. **14**, but the disclosure is not limited thereto. FIG. **14** is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure. Like elements are designated with the same reference numbers for ease of understanding and the details thereof are not repeated herein. Referring to FIG. **14**, for example, the (stacked) package structure **10J** including a circuit component **100** and the package structure **10A** disposed over the circuit component **100** is provided. The circuit component **100** may be or may include an interposer, a package substrate, a printed circuit board (PCB), a printed wiring board, and/or other carrier that is capable of carrying a wafer-form semiconductor package and providing a certain level of circuitry thereto. In the embodiments shown in FIG. **14**, the package structure **10A** is mounted on the circuit component **100** through the conductive elements **250**. In other words, the package structure **10A** is electrically coupled to the circuit component **100** through the conductive elements **250**, for example. The package structure **10A** mounted on the circuit component **100** may be substituted by one of the package structures **10B-10I** described above.

[0090] In some embodiments, an underfill **406** is formed between the gap of the circuit component **100** and the package structure **10A** to at least laterally cover the conductive elements **250**.

Alternatively, the underfill layer **406** is omitted. In one embodiment, the underfill **406** may be formed by underfill dispensing or any other suitable method. In some embodiments, a material of the underfill **406** may be the same or different from a material of the underfill **404**, the disclosure is not limited thereto. Owing to the underfill **406**, a bonding strength between the circuit component **100** and the package structure **10A** is enhanced, thereby improving the reliability of the package structure **10J**.

[0091] FIG. **15** is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure. The elements similar to or substantially the same as the elements described previously will use the same reference numbers, and certain details or descriptions (e.g. the materials, formation processes, positioning configurations, etc.) of the same elements would not be repeated herein. Referring to FIG. **14** and FIG. **15** together, the package structure **10J** depicted in FIG. **14** is similar to a (stacked) package structure **10K** depicted in FIG. **15**; the difference is that, the heat dissipating module **500A** is substituted by a heat dissipating module **500K**, where the heat dissipating module **500K** is not only bonded to the metallization layer **300** through the bonding pad **402** but also bonded to the circuit component **100** through at least one bonding pad **408**. The formation and material of the bonding pad **408** is similar to or the same as the formation and material of the bonding pad **402**, and thus are not repeated herein. The heat dissipating module **500K** include a cover **510K** and a flange **550K** connected thereto, where the cover **510K** includes a central portion **512K** and a flange portion **514K** at the periphery of the central portion **512K**, and the flange **550K** is connected to the edge of the cover **510K**. In other words, on the X-Y plane, the cover **510K** is surrounded by the flange **550K**. The flange portion

514K is between the central portion **512A** and the flange **550K** along the X-Y plane. The central portion **512K** and the flange portion **514K** are respectively similar to or the same as the central portion **512A** and the flange portion **514A** as described in FIG. 3, and thus are not repeated herein for brevity.

[0092] In some embodiments, the flange **550K** have a planar portion (not labeled) extending substantially parallel to the metallization layer **300** and a protrusion portion (not labeled) extending in a direction perpendicular to the plane defined by the planar portion. For example, as shown in FIG. 15, one end of the planar portion is connected to one end of the protrusion portion, where other end of the planar portion is connected to the flange portion **514K**, and other end of the protrusion portion is connected to the circuit component **100** through the bonding pad **408**. In some embodiments, the protrusion portion and the planar portion describe a right angle at their joint, but the disclosure is not limited thereto. In some embodiments, the protrusion portion is joined to the planar portion at different angles than 90 degrees. In some embodiments, a material of the flange **550K** is the same or similar to the material of the cover **510K**. For example, the flange **550K** and the cover **510K** are produced as a single piece (integrally formed). In an alternative embodiment, the flange **550K** and the cover **510K** are fabricated separately and then assembled to produce the heat dissipating module **500K**. Owing to the flange **550K**, the bonding strength between the metallization layer **300** and the heat dissipating module **500K** is further ensured, thereby improving the reliability of the package structure **10K**.

[0093] However, the disclosure is not limited thereto; an additional element may be adopted to secure the bonding strength between the heat dissipating module and the metallization layer to improve the reliability of the package structure in the disclosure. For illustration, a non-limiting example is provided as a package structure **10L** in FIG. 16. FIG. 16 is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure. The elements similar to or substantially the same as the elements described previously will use the same reference numbers, and certain details or descriptions (e.g. the materials, formation processes, positioning configurations, etc.) of the same elements would not be repeated herein.

[0094] Referring to FIG. 16, in some embodiments, a package structure **10L** including a supporting module **600** and the package structure **10A** disposed inside of the supporting module **600** is provided. The package structure **10A** includes the semiconductor package **200** and the heat dissipating system (having the metallization layer **300** and the heat dissipating module **500A**), where the heat dissipating system is located on and thermally coupled to the semiconductor package **200**, and the semiconductor package **200** and the heat dissipating system disposed thereon are together placed into an accommodating space in the supporting module **600**. The detail of the semiconductor package **200** has been described in FIG. 1 while the detail of the heat dissipating system (having the metallization layer **300** and the heat dissipating module **500A**) has been described in FIG. 2 through FIG. 4, and thus are not repeated herein.

[0095] For example, the supporting module **600** includes a first plate **610**, a second plate **620**, a buffer element **630**, a plurality of elastic elements **640**, a plurality of bolts **650** and a plurality of fasteners **660**, as shown in FIG. 16. In some embodiments, the buffer element **630** is sandwiched between the first plate **610** and the second plate **620** to avoid damages caused by directly physical contacts therebetween. A material of the elastic element **630** may include an elastic material capable of dissipating heat, such as a thermal foam, a thermal rubber, or a thermal cushion. In some embodiments, the bolts **650** penetrate through the first plate **610**, the second plate **620** and the buffer element **630** sandwiched thereto, and the fasteners **660** are respectively threaded onto the bolts **650** and tightened to clamp the first plate **610** and the second plate **620**. In some embodiments, the elastic elements **640** are individually located between one bolt **650** and the second plate **620** to avoid damages caused by directly physical contacts therebetween. The elastic elements **640** may be springs or the like. The fasteners **660** may be, e.g., nuts that thread to the bolts **650**. In some embodiments, the accommodating space is confined by the clamped first plate

610 and second plate **620**, where a plurality of openings (not shown) formed in the second plate **620** for allowing the pipes **710** passing through and protruding out of the supporting module **600** to facilitate flow-in or flow-out of the coolant CL, and an opening hole **610a** is formed in the first plate **610** for allowing the conductive elements **250** passing through and protruded out of the supporting module **600** for electrical connections to other external elements. For example, as shown in FIG. **16**, the conductive elements **250** are exposed through the opening **610a** of the supporting module **600**. In some embodiments, the first plate **610** is formed in a frame shape, such as a ring-shape. Owing to the supporting module **600**, an addition exerted force is applied to the package structure **10L**, the bonding strength between the metallization layer **300** and the heat dissipating module **500A** is greatly ensured, thereby improving the reliability of the package structure **10L**.

[0096] In some embodiments, as shown in FIG. **16**, the first plate **610** is located on the redistribution circuit structure **240** of the semiconductor package **200** while the second plate **620** is in contact with the heat dissipating module **500A**. For one example, the supporting module **600** is thermally coupled to and electrically isolated from the semiconductor package **200**. For another example, the supporting module **600** is thermally isolated and electrically isolated from the semiconductor package **200**. The material of the first plate **610** may be different from the material of the second plate **620**. Alternatively, the material of the first plate **610** may be the same as the material of the second plate **620**. For example, a material of the supporting module **600** includes acrylic, aluminum, stainless steel, or the like, the disclosure is not limited thereto.

[0097] In one embodiment, the material of the first plate **610** and the material of the second plate **620** are independently different from the material of the central portion **512A** of the cover **510A** and/or the material of the flange portion **514A** of the cover **510A**. In an alternative embodiment, the material of the first plate **610** and the material of the second plate **620** are independently the same or similar to the material of the central portion **512A** of the cover **510A** and/or the material of the flange portion **514A** of the cover **510A**. The disclosure is not limited thereto, as long as the supporting module **600** is electrically isolated from the semiconductor package **200**.

[0098] For example, as shown in FIG. **16**, the second plate **620**, the central portion **512A** and the flange portion **514A** are fabricated separately and then assembled together, where an additional first seal element (not shown) is further adopted between the second plate **620** and the central portion **512A** and between the second plate **620** and the flange portion **514A** to avoid damages caused by directly physical contacts therebetween. Alternatively, the second plate **620**, the central portion **512A** and the flange portion **514A** may be produced as a single piece (integrally formed), where the additional first seal element is omitted therefrom. Besides, in some embodiments, an additional second seal element (not shown) is adopted between the first plate **610** and the redistribution circuit structure **240** of the semiconductor package **200** to avoid damages caused by directly physical contacts therebetween. The additional first and second seal elements, independently, may include a seal ring made of a polymeric material or the like, the disclosure is not limited thereto.

[0099] In the disclosure, if considering the structures of the heat dissipating modules **500A** through **500F** mentioned above, the cover **510A** through the cover **510F** independently may be considered as a structure of a single piece. However, in the disclosure, a cover of the heat dissipating module may include a structure of a two-piece, see FIG. **17A** and FIGS. **19A-19B** as described below in detail.

[0100] FIG. **17A** is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure. FIG. **17B** and FIG. **17C** are enlarged and schematic cross-sectional views illustrating a configuration of a portion of the heat dissipating module depicted in FIG. **17A**. FIG. **18A** is a schematic top view illustrating the package structure depicted in FIG. **17A**, where FIG. **17A** is the cross-sectional view taken along a line BB' depicted in FIG. **18A**. FIG. **18B** and FIG. **18C** are schematic top views respectively illustrating different parts of a heat dissipating module in the package structure depicted in FIG. **17A**. Referring to FIG. **17A** and

FIG. 18A, in some embodiments, a package structure 10M includes a semiconductor package 200 and a heat dissipating system thermally coupled to the semiconductor package 200. For example, the heat dissipating system includes a metallization layer 300 (with the pillars 322 (e.g. microstructure 320) disposed thereon via the bonding pads 310) and a heat dissipating module 500M, where the metallization layer 300 is located on and thermally coupled to the semiconductor package 200, and the heat dissipating module 500M is bonded to the metallization layer 300 through the bonding pad 402 to form the heat dissipating system thermally coupled to the semiconductor package 200. The underfill 404 are formed over the metallization layer 300 exposed by the heat dissipating module 500M, where the underfill 404 surrounds the sidewall of the heat dissipating module 500M, for example. The details of the semiconductor package 200, the metallization layer 300, the bonding pads 310/402, the microstructure 320 and the underfill 404 have described in FIG. 1 through FIG. 4, and thus are not repeated herein for simplicity.

[0101] In some embodiments, the heat dissipating module 500M includes a cover 510M having a central portion 512M and a flange portion 514M at the periphery of the central portion 512M. In some embodiments, the central portion 512M includes a first portion P1 and a second portion P2 connected thereto. In some embodiments, the first portion P1 has a planar portion P1a and a protrusion portion P1b connected thereto, where the planar portion P1a extends substantially parallel to the metallization layer 300 and the protrusion portion P1b extends in a direction substantially perpendicular to a plane defined by the planar portion P1a. In some embodiments, the second portion P2 has a planar portion P2a and a protrusion portion P2b connected thereto, where the planar portion P2a extends substantially parallel to the metallization layer 300 and the protrusion portion P2b extends in a direction substantially perpendicular to a plane defined by the planar portion P2a. As shown in FIG. 17A, for example, the protrusion portion P2b of the second portion P2 is connected to the planar portion P1a of the first portion P1 through bonding pads 560 to construct the cover 510M having the structure of two-piece while the planar portion P2a of the second portion P2 is connected to the flange portion 514M.

[0102] The flange portion 514M is similar to the flange portion 514A as described in FIG. 3, and thus is not repeated herein. The materials and formations of the first portion P1 and the second portion P2 of the cover 510M are independently similar to the material and formation of the cover 510A described in FIG. 3, and thus are not repeated herein. In one embodiment, the second portion P2 and the flange portion 514M are integrally formed. In an alternative embodiment, the second portion P2 and the flange portion 514M are fabricated separately and then assembled together.

[0103] The material of the bonding pads 560 may be the same or similar to the material of the bonding pads 310 or the material of the bonding pads 402, the disclosure is not limited thereto, as long as the first portion P1 and the second portion P2 are securely and mechanically coupled to each other. Alternatively, if considering the materials of the first portion P1 and the second portion P2 are metals or metal alloys, the first portion P1 and the second portion P2 may be bonded to each other through brazing process, instead of bonding pads 560.

[0104] In some embodiments, the microstructure 320 (e.g., the pillars 322) is bonded to the metallization layer 300 through the bonding pads 310 and located inside an enclosure E2 of the heat dissipating system, where the microstructure 320 serves as part of the heat dissipating system. For example, the enclosure E2 of the heat dissipating system includes two fluidic channels 1001 and 1002 extending substantially parallel to the metallization layer 300 and spatially communicated in a vertical direction, where the pillars 322 are located in the fluidic channel 1001 confined by the metallization layer 300 and the first portion P1. The fluidic flowing channel 1002 confined by the first portion P1 and the second portion P2 may be free of the microstructures. The incident flow of the coolant CL enters the heat dissipating system via the channels 920 and passes through the fluidic flowing channel 1002 and then the fluidic channel 1001, where the incident flow is divided into different directions by the pillars 322, see the flow direction indicated by the arrows F. Owing to the two fluidic channels 1001 and 1002, the remaining time of the coolant flow inside the heat

dissipating system is increased, thereby enhancing the heat dissipating effect.

[0105] In some embodiments, the first portion **P1** is overlapped with the second portion **P2** on the X-Y plane, see the top view shown in FIG. **17A** and FIG. **18A**. For example, the first portion **P1** has a size **W1** (FIG. **18B**) less than a width **W2** (FIG. **18C**) of the second portion **P2**, where the periphery of the first portion **P1** is distant from the periphery of the second portion **P2**. In other words, a space, in an annular shape surrounding the periphery of the first portion **P1**, presents between the first portion **P1** and the flange portion **514M** to allow the coolant **CL** vertically flowing therethrough, which achieves the vertical communication between the fluidic channels **1001** and **1002**. Continued on FIG. **18B** and FIG. **18C**, the first portion **P1** includes the channel **910** (e.g., the outflow channel) while the second portion **P2** includes an opening **O1** and the channels **920** (e.g., the inflow channel) surrounding the opening **O1**, where a positioning location of the channel **910** is within a positioning location of the opening **O1**. On the X-Y plane, a size **D1** of the channel **910** is less than a size **D2** of the opening **O1**, for example. As shown in FIG. **17A**, after assembling the first portion **P1** and the second portion **P2**, a portion of the first portion **P1** (involving the channel **910**) is inserted into the opening **O1** of the second portion **P2** to facilitate the installation of the fluid circulation system **700**, in some embodiments. In some embodiments, the pipes **710** are securely installed to the channels **910**, **920** of the heat dissipating module **500M** through the washers **720**. The detail of the fluid circulation system **700** (including the pipes **710** and the washers **720**) has described in FIG. **4**, and thus is not repeated herein for simplicity.

[0106] As illustrated in FIG. **17A** and FIG. **17B** (showing an enlarged cross-sectional view indicated by a dashed box **X** in FIG. **17A**), for example, the opening of the channel **910** to the enclosure **E2** include a shape edge with an angle of 90° . In some alternative embodiments, in FIG. **17A** and FIG. **17C** (showing an enlarged cross-sectional view indicated by the dashed box **X** in FIG. **17A**), the opening of the channel **910** to the enclosure **E2** include a bevel edge **BE**. However, the disclosure is not limited thereto, the opening of the channel **910** to the enclosure **E2** may include a rounded edge (not shown). As shown in FIG. **17A**, for example, a thickness **TP1a** of the planar portion **P1a** and a thickness **TP1b** of the protrusion portion **P1b** in the first portion **P1** and a thickness **TP2a** of the planar portion **P2a** and a thickness **TP2b** of the protrusion portion **P2b** in the second portion **P2** may be selected and designated based on the demand and/or design requirement, the disclosure is not limited thereto.

[0107] FIG. **19A** and FIG. **19B** are schematic cross-sectional views illustrating a package structure in accordance with some embodiments of the disclosure. FIG. **20A** is a schematic top view illustrating the package structure depicted in FIG. **19A** and FIG. **19B**, where FIG. **19A** is the cross-sectional view taken along a line **CC'** of FIG. **20A**, and FIG. **19B** is the cross-sectional view taken along a line **DD'** of FIG. **20A**. FIG. **20B** and FIG. **20C** are schematic top views respectively illustrating different parts of a heat dissipating module in the package structure depicted in FIG. **19A** and FIG. **19B**. The elements similar to or substantially the same as the elements described above will use the same reference numbers, and certain details or descriptions of the same elements (e.g. the formations and materials) and the relationship thereof (e.g. the relative positioning configuration and electrical connection) will not be repeated herein. Referring to FIG. **17A** and FIG. **19A** through FIG. **19B** together, the package structure **10M** depicted in FIG. **17A** and a package structure **10N** depicted in FIG. **19A** and FIG. **19B** are similar; the difference is that, in the package structure **10N** of FIG. **19A** and FIG. **19B**, the heat dissipating module **500M** used to be thermally coupled to and bonded to a semiconductor package **200** is substituted by the heat dissipating module **500N**. The details of the semiconductor package **200** and the other components (e.g., the metallization layer **300**, the bonding pads **310/402**, the underfill **404**, the fluid circulation system **700** or so on) shown in FIG. **19A** and FIG. **19B** have been described in FIG. **1** through FIG. **4**, and thus are not repeated herein.

[0108] In some embodiments, the heat dissipating module **500N** includes a cover **510N** having a central portion **512N** and a flange portion **514N** at the periphery of the central portion **512N**. In

some embodiments, the central portion **512N** includes a first portion **P3** and a second portion **P4**, where the second portion **P4** has a planar portion **P4a** and a protrusion portion **P4b** connected thereto. In some embodiments, the first portion **P3** extends substantially parallel to the metallization layer **300**, while the planar portion **P4a** of the second portion **P4** extends substantially parallel to the metallization layer **300** and the protrusion portion **P4b** of the second portion **P4** extends in a direction substantially perpendicular to a plane defined by the planar portion **P4a**. As shown in FIG. **19A** and FIG. **19B**, for example, an inner side of the first portion **P3** is connected to (a sidewall of) the protrusion portion **P4b** of the second portion **P4** while an outer side of the first portion **P3** is connected to (a sidewall of) the flange portion **514N**, and the planar portion **P4a** of the second portion **P4** is connected to (one end of) the flange portion **514N** via bolts **564** to construct the cover **510N** having the structure of two-piece. The materials of the first portion **P3** and the second portion **P4** of the cover **510N** are similar to the materials of the first portion **P1** and the second portion **P2** of the cover **510A** described in FIG. **3**, and thus are not repeated herein. The flange portion **514N** is similar to the flange portion **514A** as described in FIG. **3**, and thus is not repeated herein.

[0109] In one embodiment, as shown in FIG. **19A** and FIG. **19B**, the first portion **P3** and the flange portion **514N** are integrally formed, where the first portion **P3** are bonded to (the protrusion portion **P4b** of) the second portion **P4** via an additional adhesive or by suitable technology (e.g. by brazing). In one alternative embodiment, the first portion **P3** and second portion **P4** are integrally formed, where the first portion **P3** are bonded to the flange portion **514N** via an additional adhesive or by suitable technology (e.g. by brazing).

[0110] In addition, for example, as shown in FIG. **19A** and FIG. **19B**, a buffer element **562** is sandwiched between the flange portion **514N** and the planar portion **P4a** of the second portion **P4** to avoid damages caused by directly physical contacts between the flange portion **514N** and the second portion **P4**. The buffer element **562** and the bolts **564** are respectively similar to the buffer element **630** and the bolts **650** as described in FIG. **16**, and thus are not repeated herein for brevity. Similar to the heat dissipating system depicted in FIG. **17A**, the enclosure **E2** of the heat dissipating system in the FIG. **19A** and FIG. **19B** includes two fluidic channels **1001** and **1002** extending substantially parallel to the metallization layer **300** and spatially communicated in a vertical direction, where the pillars **322** are located in the fluidic channel **1001** confined by the metallization layer **300** and the first portion **P3**. The fluidic flowing channel **1002** confined by the first portion **P3** and the second portion **P4** may be free of the microstructures. In some embodiments, the incident flow of the coolant **CL** enters the heat dissipating system via the channels **920** and passes through the fluidic flowing channel **1002** and then the fluidic channel **1001**, where the incident flow is divided into different directions by the pillars **322**, see the flow direction **F** shown in FIG. **19B**. In some embodiments, the incident flow of the coolant **CL** exists the heat dissipating system via the channels **910** by passing through the fluidic flowing channel **1001** and the fluidic channel **1002**, see the flow direction **F** shown in FIGS. **19A** and **19B**.

[0111] Continued on FIG. **19A**, FIG. **19B** and FIG. **20A**, in some embodiments, the first portion **P3** is overlapped with the second portion **P4** on the X-Y plane. For example, the first portion **P3** has a size **W3** (FIG. **20B**) less than a width **W4** (FIG. **20C**) of the second portion **P4**, where the (outer) periphery of the first portion **P3** is partially in contact with the flange portion **514N** and the (inner) periphery of the first portion **P3** is partially in contact with the second portion **P4**, while the periphery of the second portion **P4** is substantially aligned with the periphery of the flange portion **514N**. As shown in FIG. **20B** and FIG. **20C**, the first portion **P3** includes an opening **O2** with a size **D3** while the second portion **P4** includes the channels **910** (e.g., the outflow channel) and the channels **920** (e.g., the inflow channels) surrounding the channels **910**, where a positioning locations of a respective one of the channels **910** is within a positioning location of the opening **O2**, and positioning locations of rest of the channels **910** are outside of the positioning location of the opening **O2**. On the X-Y plane, the size **D3** of the opening **O2** is greater than a size **D4** of the

respective one channel **910**, for example. As shown in FIG. **19A**, after assembling the first portion **P3** and the second portion **P4**, a portion of the second portion **P4** (involving the respective one of the channels **910**) is inserted into the opening **O2** of the first portion **P3**, in some embodiments. [0112] In some embodiments, as shown in FIG. **19A**, a first space, in a discontinuous annular shape, presents between the inner periphery of the first portion **P3** and the inserted part of the second portion **P4** at the opening **O2** for allowing the coolant **CL** flowing therethrough. On the other hand, in some embodiments, as shown in FIG. **19B**, a second space, in a discontinuous annular shape, presents between the outer periphery the first portion **P3** and the flange portion **514N** for allowing the coolant **CL** flowing therethrough. Owing to such first and second spaces, the vertical communication between the fluidic channels **1001** and **1002** of the enclosure **E2** is achieved.

[0113] In the disclosure, the microstructures are enclosed inside (the fluidic channel **1001** of) the enclosure **E1** or **E2** of one heat dissipating system constituted by the metallization layer **300** and one of the heat dissipating module (e.g. **500A-500F**, **500K**, **500M-500N**). In some embodiments, the central portion **512A-512F**, **512K**, **512M-512N** may constitute the ceiling of the enclosure **E1** or **E2**, the flange portion **514A-514F**, **514K**, **514M-514N** may constitute the sidewall of the enclosure **E1** or **E2**, and the metallization layer **300** may constitute the floor of the enclosure **E1** or **E2**. For some non-limiting examples, the microstructures may be disposed on the floor of the enclosure **E1** or **E2**; see package structures **10A**, **10B**, **10C**, **10D**, **10E**, **10F**, **10G**, **10H**, **10I**, **10J**, **10K**, **10L**, **10M**, **10N** respectively depicted in FIG. **4**, FIG. **6** through FIG. **17A** and FIGS. **19A-19B**. However, the disclosure is not limited thereto; alternatively, in addition to the microstructures disposed on the floor of the enclosure **E1** or **E2**, additional microstructures may be disposed on the ceiling of the enclosure **E1** or **E2**; see package structures **100**, **10P**, **10Q**, **10R**, **10S**, **10T** and **10U** respectively depicted in FIG. **23** through FIG. **29**.

[0114] FIG. **21** through FIG. **23** are schematic cross-sectional views illustrating intermediate structures formed at various stages of a manufacturing method of a package structure in accordance with some embodiments of the disclosure, where FIG. **21** to FIG. **23** are the cross-sectional views taken along the line **AA'** of FIG. **5**. FIG. **24** through FIG. **26** are schematic cross-sectional views respectively illustrating a package structure in accordance with some embodiments of the disclosure. FIG. **27** is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure. The elements similar to or substantially the same as the elements described above will use the same reference numbers, and certain details or descriptions of the same elements (e.g. the formations and materials) and the relationship thereof (e.g. the relative positioning configuration and electrical connection) will not be repeated herein.

[0115] Referring to FIG. **21**, in some embodiments, a semiconductor package **200** is provided, and a metallization layer **300** is located over a backside surface **S200b** of the semiconductor package **200**. For example, the metallization layer **300** is in form of a blanket layer covering the semiconductor package **200**, where the metallization layer **300** is thermally coupled to and electrically isolated from the semiconductor package **200**. The detail of the semiconductor package **200** and the detail of the metallization layer **300** have been described in FIG. **1**, and thus are omitted for simplicity. In some embodiments, after forming the metallization layer **300**, bonding pads **310** and a bonding pad **402** are formed over a surface **S300** of the metallization layer **300**, where the metallization layer **300** is sandwiched between the semiconductor package **200** and the bonding pads **310**, **402**. The details of the bonding pads **310**, **402** have been described in FIG. **2**, and thus are not repeated herein. In some embodiments, after the formation of the bonding pads **310** and the bonding pad **402**, a plurality of pillars **322** are formed on the bonding pads **310** and are thermally coupled to the metallization layer **300** through the bonding pads **310**. The pillars **322** are referred to as the microstructure **320** for heat dissipating as the pillars **322** are thermally coupled to the semiconductor package **200** via the bonding pads **310** and the metallization layer **300**. The detail of the microstructure **320** (e.g., the pillars **322**) has been described in FIG. **2**, and thus is not

repeated herein.

[0116] Referring to FIG. 22, in some embodiments, a heat dissipating module **500O** is provided and bonded to the metallization layer **300** through the bonding pad **402** to form a heat dissipating system thermally coupled to and electrically isolated from the semiconductor package **200**. For example, the heat dissipating module **500O** includes a cover **510O** having a central portion **512O** and a flange portion **514O** at the periphery of the central portion **512O**. In some embodiments, as shown in FIG. 22, the heat dissipating module **500O** further includes a plurality of pillars **542** disposed on the central portion **512O** through bonding pads **530**. The central portion **512O** and the flange portion **514O** of the cover **510O** are respectively similar to or the same as the central portion **512A** and the flange portion **514A** of the cover **510A** as described in FIG. 3, and thus are not repeated herein. In some embodiments, as illustrated in FIG. 22, the pillars **542** are located on an inner surface of the central portion **512O** facing toward the metallization layer **300** and are thermally coupled to the central portion **512O** through the bonding pads **530**, where the pillars **542** are referred to as a microstructure **540**. The formation and material of the pillars **542** may be the same or similar to the formation and material of the pillars **322** as described in FIG. 2, and thus are not repeated herein. However, the disclosure is not limited thereto; alternatively, the pillars **542** may be formed by milling machining, where the pillars **542**, the central portion **512O** and the flange portion **514O** are integrally formed, and there is no bonding pad **530** between the pillars **542** and the central portion **512O**. For example, the microstructures **320** and **540** (e.g., the pillars **322**, **542**) respectively located on the metallization layer **300** and the central portion **512O** are surrounded by the flange portion **514O**, where the microstructure **320** is distant from the central portion **512O** and the microstructure **540** are distant from the metallization layer **300**. However, the disclosure is not limited thereto, the microstructure **320** (e.g., the pillars **322**) may in contact with the central portion **512O** of the cover **510O**; and/or, the microstructure **540** (e.g., the pillars **542**) may in contact with the metallization layer **300**.

[0117] In some embodiments, the heat dissipating system constituted together by the metallization layer **300** and the heat dissipating module **500O** including the cover **510O** (having the central portion **512O** and the flange portion **514O**) defines an enclosure **E1** surrounding the microstructures **320** and **540** on all sides, where a fluidic channel **1001** is confined by the enclosure **E1**. In some embodiments, the central portion **512O** includes one or more channels (e.g., **910** and **920** in the cross-sectional view of FIG. 22 and in the top view of FIG. 5) spatially communicated with the fluidic channel **1001**. In the disclosure, the channels **910**, **920** may be individually referred to as an inflow channel or an outflow channel of the enclosure **E1** included in the heat dissipating system, and may be selected and designated based on the demand and the design requirement. In some embodiments, the enclosure **E1** serves as an accommodating container of the coolant **CL** (shown in FIG. 23), where the coolant **CL** is in contact with the pillars **322**, **542** placed inside the fluidic channel **1001** to achieve heat transfer from the pillars **322**, **542** to the coolant **CL** for the purpose of heat dissipating. In some embodiments, the microstructures **320** and **540** together define a network of interstices in fluidic communication inside the enclosure **E1** of the heat dissipating system. With the microstructures **320** and **540**, the distribution of the coolant flow is further improved by increasing the vertical movement of the coolant flow inside the fluidic channel **1001**. That is, similar to the pillars **322**, the distribution of the coolant flow is improved due to the pillars **542**, thereby further enhancing the heat dissipation of the package structure **10O**.

[0118] Referring to FIG. 23, in some embodiments, a fluid circulation system **700** is installed on the heat dissipating module **500O** in correspondence with the channels **910**, **920** to facilitate flowing of the coolant **CL** in and out of the enclosure **E1** of the heating dissipating system. The detail of the fluid circulation system **700** has discussed in FIG. 4, and thus is omitted for brevity. Up to here, the package structure **10O**, which is installed with the heat dissipating system having the heating dissipating module **500O** with the microstructure **540** disposed thereon and the metallization layer **300** with the microstructure **320** disposed thereon, is manufactured. In the

disclosure, some of the microstructure **320** (e.g., pillars **322**) located on the metallization layer **300** are positioned underneath the openings of the channels **920**, where the coolant CL flows into the heat dissipating system through the channels **920** and is further divided into sub-flows of different directions due to the presence of such microstructure **320**. Owing to the microstructure **320** positioned underneath the openings of the channels in the heat dissipating system, the distribution of the coolant flow is improved, thereby enhancing the heat dissipation of the package structure **100**. It is appreciated that the number of the microstructures (e.g., **322**, **324**, **326**, **328**) in the heat dissipating system may not be affected by the presence of the microstructures (e.g., **542**, **544**, **546**, **548**), where the number of the microstructures (e.g., **322**, **324**, **326**, **328**) and the number of the microstructures (e.g., **542**, **544**, **546**, **548**) are independently selected and designated based on the demand and design layout, as long as there is at least one micro-pillar (e.g., **322** or **324**) disposed at a location corresponding to the opening of each of the inflow/outflow channels of the heat dissipating system for splitting the flow.

[0119] The modifications to the heat dissipating module discussed in FIG. **6** through FIG. **10** and/or the modifications to the microstructures discussed in FIG. **11** through FIG. **13** may be adopted in the heat dissipating module **500O** and/or the microstructure **320** of the package structure **100** depicted in FIG. **23**. In addition, similar modifications to the microstructure **320** discussed in FIG. **11** through FIG. **13** may be also adopted in the microstructure **540** of the package structure **100** depicted in FIG. **23** (e.g., FIG. **24** through FIG. **26**, which will be further described later in detail). The elements similar to or substantially the same as the elements described above will use the same reference numbers, and certain details or descriptions of the same elements (e.g. the formations and materials) and the relationship thereof (e.g. the relative positioning configuration and electrical connection) will not be repeated herein.

[0120] For example, referring to FIG. **23** and FIG. **24** together, the package structure **100** of FIG. **23** is similar to a package structure **10P** of FIG. **24**; the difference is that, the package structure **10P** of FIG. **24** adopts a heat dissipating module **500P** for substituting the heat dissipating module **500O**. The central portion **512P** and the flange portion **514P** of the cover **510P** included in the heat dissipating module **500P** are respectively similar to or the same as the central portion **512O** and the flange portion **514O** of the cover **510O** included in the heat dissipating module **500O** as described in FIG. **23**, and thus are not repeated herein for brevity. In some embodiments, for the heat dissipating module **500P** of the package structure **10P**, the microstructure **540** is substituted by a microstructure **540A**, where the microstructure **540A** includes the pillars **542** and splitter **544** disposed on the central portion **512P** through the bonding pads **530**. The formation and material of the splitters **544** are the same or similar to the formation and material of the splitters **324**, and thus are not repeated herein. However, the disclosure is not limited thereto; alternatively, the splitters **544** may be formed by milling machining, where the splitters **544**, the central portion **512P** and the flange portion **514P** are integrally formed, and there is no bonding pad **530** between the splitters **544** and the central portion **512P**. Similar to the splitters **324**, the distribution of the coolant flow is further improved due to the splitters **544**, thereby further enhancing the heat dissipation of the package structure **10P**.

[0121] However, the disclosure is not limited thereto. The microstructure **540** may be substituted by a microstructure **540B** (e.g. a heat dissipating module **500Q** of a package structure **10Q** as shown in FIG. **25**) or substituted by a microstructure **540C** (e.g. a heat dissipating module **500R** of a package structure **10R** as shown in FIG. **26**).

[0122] For one example, referring to FIG. **23** and FIG. **25** together, the package structure **100** of FIG. **23** is similar to a package structure **10Q** of FIG. **25**; the difference is that, the package structure **10Q** of FIG. **25** adopts a heat dissipating module **500Q** for substituting the heat dissipating module **500O**. The central portion **512Q** and the flange portion **514Q** of the cover **510Q** included in the heat dissipating module **500Q** are respectively similar to or the same as the central portion **512O** and the flange portion **514O** of the cover **510O** included in the heat dissipating

module **500O** as described in FIG. 23, and thus are not repeated herein for brevity. In some embodiments, for the heat dissipating module **500Q** of the package structure **10Q**, the microstructure **540** is substituted by the microstructure **540B**, where the microstructure **540B** includes the pillars **542** and baffles **546** disposed on the central portion **512Q** through the bonding pads **530**. The formation and material of the baffles **546** are the same or similar to the formation and material of the baffles **326**, and thus are not repeated herein. However, the disclosure is not limited thereto; alternatively, the baffles **546** may be formed by milling machining, where the baffles **546**, the central portion **512Q** and the flange portion **514Q** are integrally formed, and there is no bonding pad **530** between the baffles **546** and the central portion **512Q**. Similar to the baffles **326**, the movement of the coolant flow is controllable by the baffle **546**, which leads to a fast heat dissipation, thereby further enhancing the heat dissipation of the package structure **10Q**.

[0123] For another example, referring to FIG. 23 and FIG. 26 together, the package structure **10O** of FIG. 23 is similar to a package structure **10R** of FIG. 26; the difference is that, the package structure **10R** of FIG. 26 adopts a heat dissipating module **500R** for substituting the heat dissipating module **500O**. The central portion **512R** and the flange portion **514R** of the cover **510R** included in the heat dissipating module **500R** are respectively similar to or the same as the central portion **512O** and the flange portion **514O** of the cover **510O** included in the heat dissipating module **500O** as described in FIG. 23, and thus are not repeated herein for brevity. In some embodiments, for the heat dissipating module **500R** of the package structure **10R**, the microstructure **540** is substituted by the microstructure **540C**, where the microstructure **540C** includes the pillars **542** and fins **548** disposed on the central portion **512R** through the bonding pads **530**. The formation and material of the fins **548** are the same or similar to the formation and material of the fins **328**, and thus are not repeated herein. However, the disclosure is not limited thereto; alternatively, the fins **548** may be formed by milling machining, where the fins **548**, the central portion **512R** and the flange portion **514R** are integrally formed, and there is no bonding pad **530** between the fins **548** and the central portion **512R**. Similar to the fins **328**, the movement of the coolant flow is further controllable due to the fins **548**, which leads to a fast heat dissipation, thereby further enhancing the heat dissipation of the package structure **10R**.

[0124] It is appreciated that in the disclosure, the microstructures (e.g., **542**, **544**, **546**, **548**) may be, in part or all, integrally formed with the cover of a heat dissipating module. The disclosure is not limited thereto.

[0125] In some embodiments, the flow movements of the coolant CL due to the presences of the microstructures (e.g., **542**, **544**, **546**, **548**) inside the heat dissipating system are demonstrated in FIG. 33 for illustrative purposes, where an arrangement of the microstructures (e.g., **542**, **544**, **546**, **548**) is not specifically limited thereto. The non-limiting examples of possible arrangements of the microstructures (e.g., **542**, **544**, **546**, **548**) will be described in FIG. 34 through FIG. 45 later for further details. Referring to FIG. 33, in some embodiments, if assuming the coolant CL proceeds from the right side of the drawing towards the left end of the drawing (along the direction X), the incident flow (indicated by arrows) is deviated by the pillars **542** and the splitters **544**, forced to pass under the fins **548**, and deviated by the baffle **546**. In some embodiments, the incident flow flows in a space between two baffles **546**, between two fins **548** or between one baffle **546** and one fin **548**, and passes under the pillars **542** and the splitters **544**. As show in FIG. 33, the microstructures (e.g., **542**, **544**, **546**, **548**) are separated from each other on the X-Y plane, for example. In other words, the microstructures (e.g., **542**, **544**, **546**, **548**) are not overlapped to each other in direction Z. In some embodiments, the microstructures (e.g., **542**, **544**, **546**, **548**) may be coated with a thermally conductive material. In some embodiments, the coating material includes graphene or diamond-like carbon (DLC). In some embodiments, the coating material includes metals such as copper or aluminum.

[0126] Furthermore, in the above non-limiting examples, along the direction Z, the thicknesses of the pillars **322** are the same to each other while the thicknesses of the pillars **542** are also the same

to each other. However, the disclosure is not limited thereto; alternatively, the thicknesses of the pillars **322** may be different, in part or all. Similarly, the thicknesses of the pillars **542** may be different, in part or all. In the disclosure, such thickness modification in pillars **322/542** may also be applied to the thicknesses of the splitters **324/544** and/or the thicknesses of the fins **328/548**.

[0127] In the disclosure, the microstructures may be further located on the heat dissipating module in the heat dissipating system having the structure of a two-piece, besides the microstructures located on the metallization layer. The microstructures (e.g., **542**, **544**, **546**, **548**) may further be introduced to the package structure **10M** of FIG. **17A**, see FIG. **27**. For example, referring to FIG. **17A** and FIG. **27** together, the package structure **10M** of FIG. **17A** is similar to a package structure **10S** of FIG. **27**; the difference is that, the package structure **10U** of FIG. **27** adopts a heat dissipating module **500S** for substituting the heat dissipating module **500M**. The central portion **512S** and the flange portion **514S** of the cover **510S** included in the heat dissipating module **500S** are respectively similar to or the same as the central portion **512M** and the flange portion **514M** of the cover **510M** included in the heat dissipating module **500M** as described in FIG. **17A**, and thus are not repeated herein for brevity. In some embodiment, the heat dissipating module **500S** further includes a plurality of pillars **542** disposed on the first portion **P1** of the central portion **512S** of the cover **510S** through a plurality of bonding pads **530**. With the pillars **322** and **542**, the distribution of the coolant flow is further improved by increasing the vertical movement of the coolant flow inside the fluidic channel **1001**. The details of the rest components included in the package structure **10S** of FIG. **27** are the same as the components included in the package structure **10M** of FIG. **17A**, and thus are not repeated herein for simplicity. Furthermore, the modification to the package structure **10N** of FIGS. **19A-19B** (e.g., having the heat dissipating module with a cover having the two-piece structure) is very similar to the modification of the package structure **10M** of FIG. **17A** discussed above, and thus are omitted herein. Similarly, the microstructures (e.g., **542**, **544**, **546**, **548**) may further be introduced to package structures **10A**, **10B**, **10C**, **10D**, **10E**, **10F**, **10G**, **10H**, **10I**, **10J**, **10K**, **10L** (e.g., having the heat dissipating module with a cover having the single-piece structure) respectively depicted in FIG. **4** and FIG. **6** through FIG. **16**.

[0128] FIG. **28A** through FIG. **28B** are schematic cross-sectional views respectively illustrating the configurations of microstructures of a heat dissipating system in a package structure in accordance with some embodiments of the disclosure. FIG. **29A** through FIG. **29B** are schematic cross-sectional views respectively illustrating the configurations of microstructures of a heat dissipating system in a package structure in accordance with some embodiments of the disclosure. The elements similar to or substantially the same as the elements described above will use the same reference numbers, and certain details or descriptions of the same elements (e.g. the formations and materials) and the relationship thereof (e.g. the relative positioning configuration and electrical connection) will not be repeated herein. It is appreciated that the semiconductor package **200** depicted in FIG. **28A**, FIG. **28B**, FIG. **29A** and FIG. **29B** are illustrated in a simplified way, where the components included in the semiconductor package **200** are omitted for simplicity. The detail of the semiconductor package **200** is described in FIG. **1**, and thus is not repeated herein.

[0129] In some embodiments, if considering the semiconductor package **200** of a package structure undergoes a warpage caused by tensile stress, a profile of a vertical cross-section of the pillars **322** located on metallization layer **300** is in a form of a smiling pattern. Referring to FIG. **23** and FIG. **28A** together, the package structure **10O** of FIG. **23** is similar to a package structure **10T** of FIG. **28A**; the difference is that, the package structure **10T** of FIG. **28A** adopts a heat dissipating module **500T** for substituting the heat dissipating module **500O**. The central portion **512T** and the flange portion **514T** of the cover **510T** included in the heat dissipating module **500T** are respectively similar to or the same as the central portion **512O** and the flange portion **514O** of the cover **510O** included in the heat dissipating module **500O** as described in FIG. **23**, and thus are not repeated herein for brevity. In some embodiments, for the heat dissipating module **500T** of the package structure **10T**, the microstructure **540** (e.g. the pillars **542**) is only arranged on the central portion

512T at an edge region **ER** of the enclosure **E1** through the bonding pads **530**, and the microstructure **320** (e.g. the pillars **322**) is only arranged on the metallization layer **300** through the bonding pads **310** at a central region **CR** of the enclosure **E1**.

[0130] Due to the warpage caused by tensile stress, the edge of the semiconductor package **200** is bent upward along the direction **Z** as illustrated in FIG. **28A**, where the pillars **542** at the edge region **ER** of the enclosure **E1** are in contact with the metallization layer **300** through the bonding pads **310**, and the pillars **322** at the central region **CR** of the enclosure **E1** are spacing away from the central portion **512T** with different distances, for example. As shown in FIG. **28A**, for example, at the central region **CR**, the spacing distances between the pillars **322** and the central portion **512T** are increased gradually from a periphery of the central region **CR** towards to a center of the central region **CR**. Alternatively, the bonding pads **310** between the pillars **542** and the metallization layer **300** may be omitted.

[0131] However, the disclosure is not limited thereto; alternatively, some of the pillars **542** at the edge region **ER** of the enclosure **E1** may be spacing away from the metallization layer **300**, see FIG. **28B**. For example, as shown in FIG. **28B**, at the edge region **ER**, the spacing distances between the pillars **542** and the metallization layer **300** (e.g. the bonding pads **310** located thereon) are increased gradually from an outer periphery of the edge region **ER** towards to an inner periphery of the edge region **ER**. That is, in other word, at the edge region **ER**, the pillars **542** located near the outer periphery are in contact with the metallization layer **300** while the pillars **542** located near the inner periphery are spacing away from the metallization layer **300**. In a further alternative embodiment, besides the pillars **322** located underlying (or immediately underneath) the channels **910** and **920**, the rest of the pillars **322** arranged in the central region **CR** are in contact with the central portion **512T** through the bonding pads **530**.

[0132] In alternative embodiments, if considering the semiconductor package **200** of a package structure undergoes a warpage caused by compressive stress, a profile of a vertical cross-section of the pillars **322** located on metallization layer **300** is in a form of a crying pattern. Referring to FIG. **23** and FIG. **29A** together, the package structure **10O** of FIG. **23** is similar to a package structure **10U** of FIG. **29A**; the difference is that, the package structure **10U** of FIG. **29A** adopts a heat dissipating module **500U** for substituting the heat dissipating module **500O**. The central portion **512U** and the flange portion **514U** of the cover **510TU** included in the heat dissipating module **500U** are respectively similar to or the same as the central portion **512O** and the flange portion **514O** of the cover **510O** included in the heat dissipating module **500O** as described in FIG. **23**, and thus are not repeated herein for brevity. In some embodiments, for the heat dissipating module **500U** of the package structure **10U**, the microstructure **540** (e.g. the pillars **542**) is only arranged on the central portion **512U** at an edge region **ER** of the enclosure **E1** through the bonding pads **530**, and the microstructure **320** (e.g. the pillars **322**) is only arranged on the metallization layer **300** through the bonding pads **310** at a central region **CR** of the enclosure **E1**.

[0133] Due to the warpage caused by compressive stress, the edge of the semiconductor package **200** is bent downward along the direction **Z** as illustrated in FIG. **29A**, where the pillars **542** at the edge region **ER** of the enclosure **E1** are in contact with the metallization layer **300** through the bonding pads **310**, and the pillars **322** at the central region **CR** of the enclosure **E1** are spacing away from the central portion **512U** with different distances, for example. As shown in FIG. **29A**, for example, at the central region **CR**, the spacing distances between the pillars **322** and the central portion **512U** are decreased gradually from a periphery of the central region **CR** towards to a center of the central region **CR**. However, the disclosure is not limited thereto. Alternatively, the bonding pads **310** between the pillars **542** and the metallization layer **300** may be omitted.

[0134] However, the disclosure is not limited thereto; some of the pillars **542** at the edge region **ER** of the enclosure **E1** may be spacing away from the metallization layer **300**, see FIG. **29B**. For example, as shown in FIG. **29B**, at the edge region **ER**, the spacing distances between the pillars **542** and the metallization layer **300** (e.g. the bonding pads **310** located thereon) are increased

gradually from an outer periphery of the edge region ER towards to an inner periphery of the edge region ER. That is, in other word, at the edge region ER, the pillars **542** located near the outer periphery are in contact with the metallization layer **300** while the pillars **542** located near the inner periphery are spacing away from the metallization layer **300**. In a further alternative embodiment, besides the pillars **322** located underlying (or immediately underneath) the channels **910** and **920**, the rest of the pillars **322** arranged in the central region CR are in contact with the central portion **512T** through the bonding pads **530**.

[0135] FIG. **30** is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure. FIG. **31A** is a schematic three-dimensional side-view diagram of a heat dissipating module in a package structure in accordance with some embodiments of the disclosure. FIG. **31B** is a schematic top view illustrating the heat dissipating module depicted in FIG. **31A**. FIG. **31C** is a schematic bottom view illustrating the heat dissipating module depicted in FIG. **31A**. FIG. **32** is a schematic cross-sectional view illustrating a package structure in accordance with some embodiments of the disclosure. The elements similar to or substantially the same as the elements described above will use the same reference numbers, and certain details or descriptions of the same elements (e.g. the formations and materials) and the relationship thereof (e.g. the relative positioning configuration and electrical connection) will not be repeated herein.

[0136] Referring to FIG. **30**, in some embodiments, a package structure **10V** includes a semiconductor package **200**, a heat dissipating system thermally coupled to the semiconductor package **200**, and a circuit component **100** bonded to the semiconductor package **200**, where the semiconductor package **200** is electrically coupled to the circuit component **100** and is between the circuit component **100** and the heat dissipating system. For example, as shown in FIG. **30**, a underfill **406** is formed between the gap of the circuit component **100** and the semiconductor package **200** to further secure the bonding strength between the circuit component **100** and the semiconductor package **200**. The detail of the semiconductor package **200** has described in FIG. **1**, the details of the circuit component **100** and the underfill **406** have described in FIG. **14**, and thus are not repeated herein.

[0137] For example, the heat dissipating system includes a metallization layer **300**, a heat dissipating module **500V**, and a plurality of connectors **800** connecting the metallization layer **300** and the heat dissipating module **500V** through bonding pads **810** and **820**. In some embodiments, the metallization layer **300** is located on and thermally coupled to the semiconductor package **200**, and is electrically isolated from the semiconductor package **200**. The detail of the metallization layer **300** has described in FIG. **1**, and thus is not repeated herein. In some embodiments, the connectors **800** are bonded to and thermally coupled to the metallization layer **300** through the bonding pads **810**, and are bonded to and thermally coupled to the heat dissipating module **500V** through the bonding pad **820**. The connectors **800** may be referred to as conductive connectors having capable of heat transfer for thermal dissipation, in the disclosure. For example, the connectors **800** are conductive joints made of metal (such as copper, aluminum, etc.), metal alloy, a thermal adhesive material, or the like. In some embodiments, the formations and materials of the bonding pads **810**, **820** are the same or similar to the formations and materials of the bonding pads **310**, **530**, and thus are not omitted herein. For example, the bonding pads **810** are formed on the metallization layer **300** and the bonding pads **820** are formed on a surface **S4** of the heat dissipating system, where the connectors **800** are bonded to one of the bonding pads **810** and the bonding pads **820** first and then bonded to other one of the bonding pads **810** and the bonding pads **820** by soldering process or the like. After bonding, in some embodiments, the heat dissipating module **500V** is electrically isolated and thermally coupled to the semiconductor package **200** through the metallization layer **300**, the connectors **800** and the bonding pads **810**, **820**.

[0138] For example, the heat dissipating module **500V** includes a hollow block **510V** having a central portion **512V**, a flange portion **514V** and a backside portion **516V**, where the central portion **512V** is located on and overlapped with the backside portion **516V** along the direction Z, and the

flange portion **514V** connecting the periphery of the central portion **512V** and the periphery of the backside portion **516V**, as shown in FIG. **30** in further view of FIG. **31B** and FIG. **31C**. For example, as shown in FIG. **30** and FIG. **31A** through FIG. **31C**, an enclosure **E1** is defined by the central portion **512V**, the flange portion **514V** and the backside portion **516V**, where a fluidic channel **1001** is confined by the enclosure **E1**. In the heat dissipating system, the central portion **512V** may constitute the ceiling of the enclosure **E1**, the flange portion **514V** may constitute the sidewall of the enclosure **E1**, and the backside portion **516V** may constitute the floor of the enclosure **E1**, where a microstructure **320** (e.g., a plurality of pillars **322**) is located in the fluidic channel **1001** on the backside portion **516V** through bonding pads **310** and surrounded by the enclosure **E1**. In some embodiments, the central portion **512V** includes channels **910**, **920** (FIG. **31A** and FIG. **31B**) spatially communicated with the enclosure **E1**.

[0139] In some embodiments, the enclosure **E1** serves as an accommodating container of a coolant **CL**, where the coolant **CL** is in contact with the microstructure **320** (pillars **322**) placed inside the fluidic channel **1001** to achieve heat transfer from the pillars **322** to the coolant **CL** for the purpose of heat dissipating. In the disclosure, the microstructure **320** (e.g., some of the pillars **322**) located on the backside portion **516V** are positioned underneath the openings of the channels **920**, where the coolant **CL** flows into the heat dissipating system through the channels **920** and is further divided into sub-flows of different directions due to the presence of such microstructure **320**. Owing to the microstructure **320** positioned underneath the openings of the channels in the heat dissipating system, the distribution of the coolant flow is improved, thereby enhancing the heat dissipation of the package structure **10V**. In alternative embodiments, additional microstructures (e.g., **324**, **326**, **328** and/or **542**, **544**, **546**, **548**) are further adopted in the heat dissipating system. In some embodiments, a fluid circulation system **700** is installed onto the heat dissipating module **500V**, where the fluid circulation system **700** includes pipes **710** and washers **720**, the pipes **710** are securely installed to the channels **910**, **920** of the heat dissipating module **500V** through the washers **720**. In addition, the detail of the fluid circulation system **700** is discussed in FIG. **4**, and thus is not repeated herein for brevity.

[0140] The formations and materials of the central portion **512V** and the flange portion **514V** may be similar to the formations and materials of the central portion **512A** and the flange portion **514A** as described in FIG. **3**, and thus are not repeated herein. In some embodiments, the formation and material of the backside portion **516V** are similar to the formations and materials of the central portion **512A** and the flange portion **514A**. For example, a thickness **T516** of the backside portion **516V** may be selected and designated based on the demand and/or design requirement, the disclosure is not limited thereto. For example, the central portion **512V**, the flange portion **514V** and the backside portion **516V** are integrally formed. In an alternative embodiment, the backside portion **516V** and the central and flange portions **512V**, **514V** are fabricated separately and then assembled to produce the hollow block **510V**. The disclosure is not limited thereto.

[0141] However, the disclosure is not limited thereto, the inflow/outflow channels (e.g., **910**, **920**), in part or all, may be relocated. A non-limiting example is shown as a package structure **10X** of FIG. **32**, where the package structure **10X** of FIG. **32** is similar to the package structure **10V** of FIG. **30**, except the locations of the inflow channels such as **920**. Referring to FIG. **32**, the package structure **10X** includes a heat dissipating module **500X**, where the heat dissipating module **500X** includes a hollow block **510X** having a central portion **512X**, a flange portion **514X** and a backside portion **516X**, where the flange portion **514X** includes at least the channels **920** while the central portion **512X** includes at least the channel **910**. As shown in FIG. **32**, the coolant **CL** flowing into the heat dissipating system through the channels **920** are still divided into sub-flows of different directions (see the flow directions **F**) due to the presence of the microstructure **320** (e.g., pillars **322**). The formations and materials of the central portion **512X**, the flange portion **514X** and the backside portion **516X** are similar to the central portion **512V**, the flange portion **514V** and the backside portion **516V** as described in FIG. **30**, and thus are not repeated herein.

[0142] FIG. 34 through FIG. 45 are schematic top views respectively illustrating an arrangement of microstructures and inflow/outflow channels of a heat dissipating system in a package structure in accordance with some embodiments of the disclosure. The elements similar to or substantially the same as the elements described above will use the same reference numbers, and certain details or descriptions of the same elements (e.g. the formations and materials) and the relationship thereof (e.g. the relative positioning configuration and electrical connection) will not be repeated herein. In FIG. 34 through FIG. 45, non-limiting examples of possible arrangements of the microstructures (e.g., 546, or 548) and the inflow/outflow channels (e.g., Ip (indicating the inflow channels) and Op (indicating the outflow channels)) are provided for illustrative purposes, and are not intended to limit the disclosure. It is appreciated that the inflow channels Ip and the outflow channels Op can be modified (e.g. switching the locations, the number, the size, and so on in the way of which the channels 910, 920 may adopted as described above); the disclosure is not limited to the non-limiting examples illustrated herein. Although the pillars 542 and the splitters 544 are not shown in FIG. 34 through FIG. 45 for simplicity, one or more pillars 542 (not shown) and/or one or more splitter 544 (not shown) may be formed on the central portion 512 between two fins 548, between two baffles 546 and/or between one fin 548 and one baffle 546 as long as the microstructures (e.g. 322 and/or 324) located on the metallization layer or the backside portion of the heat dissipating module are positioned corresponding to (e.g. underneath) the openings of the inflow/outflow channels. The numbers of the splitters 544 and the pillars 542 are not limited to the disclosure, and may be selected based on the demand and the design layout.

[0143] Referring to FIG. 34, in some embodiments, the microstructures (e.g., 542, 544, 546, or 548) and inflow/outflow channels (e.g., Ip and Op) are located on the central portion 512, where an outflow channel Op is located at the center of the central portion 512 while inflow channels Ip are located at the periphery of the central portion 512. With such arrangement, the flow direction F of the coolant (not shown) is from the periphery of the central portion 512 toward the center of the central portion 512. For example, there are one outflow channel Op and eight inflow channels Ip shown in FIG. 34 for illustrative purposes, however the numbers of the inflow/outflow channels Ip and Op may not be limited thereto. The outflow channel Op may be one or more than one while the inflow channels Ip may be one or more than one. Alternatively, the outflow channel Op may be replaced by one inflow channel Ip, and/or the inflow channels Ip, in part or all, may individually be replaced by one outflow channel Op.

[0144] In some embodiments, as illustrated in FIG. 34, a plurality of baffles 546 are located on the central portion 512 and connected to the flange portion 514 at the periphery of the central portion 512, where the baffles 546 individually extend over the central portion 512 from the periphery of the central portion 512 to the edge of the outflow channel Op located at the center of the central portion 512 so as to divide the central region 512 into several partition regions (not labeled) for better controlling the coolant flow. For example, as shown in FIG. 34, the central portion 512 (in a solid circle shape) is divided into four equal sections (e.g., the partition regions in arc shapes), where each of the sections has two inflow channels Ip, and all the sections share the one same outflow channel Op. However, the disclosure is not limited thereto, the number of the partition regions is not limited in the disclosure, and may be adjusted based on the demand and design layout. Alternatively, there may be no partition region (FIG. 41), two partition regions (FIG. 40), three partition regions, four partition regions (FIG. 35, FIG. 36, FIG. 39, FIG. 43), five partition regions (FIG. 42), six partition regions, seven partition regions, eight partition regions (FIG. 37, FIG. 38, FIG. 44, FIG. 45), and so on. In one embodiment, the areas of the partition regions are the same. Alternatively, the areas of the partition regions are different, in part or all.

[0145] In alternative embodiments, one or more fins 548 may be formed on the central portion 512 with the partition regions (FIG. 35-FIG. 39, FIG. 42-FIG. 45) to facilitate the control in the flow direction of the coolant. The numbers of the fins 548 and the baffles 546 are not limited to the disclosure, and may be selected based on the demand and the design layout.

[0146] In some embodiments, a least one recess **R1** may be formed in the central portion **512**. For example, as shown in FIG. **34**, there are four recesses **R1** formed in the central portion **512**, where each of the recesses **R1** is located in one of the four equal sections, and in each section, a positioning location of the recess **R1** is overlapped with positioning locations of the two inflow channels **Ip**. Alternatively, the recess **R1** may be only overlapped with one of the two inflow channels **Ip**; or, more recesses **R1** may be formed in the central portion **512**, where each recess **R1** may be only overlapped with each inflow channel **Ip**. In some embodiments, the recesses **R1** may be omitted or be, in part or all, substituted by a recess **R4**. The disclosure is not limited thereto. In other embodiments, one recess **R3** may be formed in the central portion **512** and a positioning location of the recess **R3** is overlapped with a positioning location of the outflow channels **Op**. Alternatively, the recess **R3** may be substituted by the recess **R4**. The details of the recesses **R1**, **R3** and **R4** have been described in FIG. **3** and FIG. **6**-FIG. **10**, and thus are not repeated herein.

[0147] Referring to FIG. **35**, in some embodiments, there are eight baffles **546** formed on the central portion **512**, where the baffles **546** extends from the center toward the periphery of the central portion **512**, four of the baffles **546** are each connected to a partition **546p** located at the periphery of the central portion **512** to form a partition wall **PW** for dividing the central portion **512** into four equal sections, and the other four baffles **546** are each divides one section into two parts. The partitions **546p** are connected to the flange portion **514** at the periphery of the central portion **512**, for example. In some embodiments, there are three fins **548** in each part of the above equal sections to facilities the control in the flow direction of the coolant. However, the number of the fins **548** and baffles **546** are not limited in the disclosure, and may be selected or designed based on demand and design layout. In addition, for example, the inflow channel **Ip** is formed in the central portion **512** at the center thereof, and four outflow channels **Op** are formed in the central portion **512** at the periphery thereof, where one outflow channel **Op** is placed into one section. That is, in some embodiments, each of the sections has one outflow channel **Op**, and all the sections share the one same inflow channel **Ip**.

[0148] Referring to FIG. **36**, in some embodiments, there are four partition walls for dividing the central portion **512** into four equal sections, where each section is further divided into four parts by three additional baffles **546**, and there are three fins **548** in each part of the above equal sections to facilities the control in the flow direction of the coolant. However, the number of the fins **548** and baffles **546** are not limited in the disclosure, and may be selected or designed based on demand and design layout. In some embodiments, the inflow channel **Ip** is formed in the central portion **512** at the center thereof, and four outflow channels **Op** are formed in the central portion **512** at the periphery thereof, where one outflow channel **Op** is placed into one section. That is, in some embodiments, each of the sections has one outflow channel **Op**, and all the sections share the one same inflow channel **Ip**.

[0149] Referring to FIG. **37** through FIG. **39**, in some embodiments, there are eight partition walls for dividing the central portion **512** into eight equal sections, where there are three fins **548** in each of the above equal sections to facilities the control in the flow direction of the coolant. However, the number of the fins **548** and baffles **546** are not limited in the disclosure, and may be selected or designed based on demand and design layout. In some embodiments, the inflow channel **Ip** is formed in the central portion **512** at the center thereof, and eight outflow channels **Op** are formed in the central portion **512** at the periphery thereof, where one outflow channel **Op** is placed into one section. That is, in some embodiments, each of the sections has its one outflow channel **Op**, and all the sections share the one same inflow channel **Ip**. However, the disclosure is not limited thereto; in alternative embodiments, a fin **548**, which is formed in form of a ring (e.g., a fin ring **548r**), is formed on the central portion **512** in a concentric manner with the inflow channel, and is connected with the partition walls **PW** by separating the fins **548** and the outflow channels at its two opposite sides, see FIG. **38**. In further alternatively, the recesses **R1** are formed in the central region **512**, where a positioning location of each recess **R1** is overlapped with a positioning location of each

outflow channel Op on the X-Y plane, as shown in FIG. 39.

[0150] Referring to FIG. 40, in some embodiments, two baffles 546 are formed on the central portion 512 in a spiral shape and are aligned (e.g., connected to) at a center of the central portion 512, where the center of the central portion 512 is referred to as an intersected location of the baffles 546. For example, the baffles 546 spiral outwards in opposite directions from their intersected location. After spiraling outward, two independent fluidic channels are confined by the baffles 546, and each of the fluidic channels has one inflow channel Ip and one outflow channel Op. In some embodiments, the fluidic channels independently have constant widths. However, the disclosure is not limited thereto; alternatively, the fluidic channels have, in part or all, non-constant widths. In some embodiments, the widths of the fluidic channels are different from one another.

[0151] Referring to FIG. 41, in some embodiments, a plurality of recesses R4 are formed in the central portion 512, where the recesses R4 are in annular shapes with different sizes and are arranged in a concentric manner. For example, there are two recesses R4, where one inner recess R4 has an annular shape close to the center of the central portion 512 while one outer recess R4 has an annular shape recess close to the periphery of the central portion 512 and enclosing the inner recess R4, as shown in FIG. 41. In some embodiments, one outflow channel Op is placed at the center of the central portion 512, four inflow channels Ip are placed at the inner recess R4, and four additional outflow channels Op are placed at the outer recess R4. For example, the two opposite inflow channels Ip and the center outflow channel Op are substantially aligned along a diameter of the central portion 512 but not aligned with any outflow channels Op at the outer recess R4. On the other hand, the two opposite outflow channels Op at the outer recess R4 and the center outflow channel Op are substantially aligned along a diameter of the central portion 512 but not aligned with any inflow channels Ip at the inner recess R4.

[0152] Referring to FIG. 42, in some embodiments, a baffle 546, which is formed in form of a ring (e.g., a baffle ring 546r), is formed on the central portion 512 in a concentric manner to have two independent partition regions (e.g. an inner region and an outer region surrounding the inner region). In some embodiments, within the inner region, one inflow channel Ip is formed in the central portion 512 at the center of the inner region and four outflow channels Op are formed in the central portion 512 at the periphery of the inner region. For example, the fins 548 are formed on the central portion 512 with the inner region, and extend from the inflow channel Ip toward the outflow channels Op in a spiral manner (e.g., clockwise). Alternatively, the fins 548 may extend counterclockwise. In some embodiments, within the outer region, four additional baffles 546 are formed on the central portion 512 and connected to the baffle ring 546r, where the baffles extend from the baffle ring 546r toward the periphery of the central portion 514 in a spiral manner (e.g., clockwise) to further divide the outer region into four equal sections. For example, four inflow channels Ip are formed in the central portion 512 at the joints of the baffles 546 and the baffle ring 546r, respectively; and four outflow channels Op are formed in the central portion 512 at the periphery thereof. For example, additional three fins 548 are formed on the central portion 512 in each equal section, and extend from the inflow channel Ip toward the outflow channels Op in a spiral manner (e.g., clockwise). Alternatively, the additional baffles 546 and/or the additional fins 548 may extend counterclockwise, the disclosure is not limited thereto.

[0153] Referring to FIG. 43, in some embodiments, an inflow channel Ip is formed in the central portion 512 at the center thereof, and four baffles 546 are formed on the central portion 512 and connected to the edge of the inflow channel Ip, where the baffles 546 extend from the inflow channel Ip to the periphery of the central portion 512 (to connect to the flange portion 514) in a spiral manner (e.g., clockwise), and the central portion 512 is divided into four equal sections. For example, each of the baffles 546 may include at least one bent portion due to extending in the spiral manner. In some embodiments, additional inflow channels Ip are formed in the central portion 512 at the bent portion of each baffle 546, and four outflow channels Op are formed in the central portion 512 at the joints of baffles 546 and the flange portion 514. With such, each section has its

own one inflow channel Ip and one outflow channel Op while all the sections share extra one same inflow channel Ip. In some embodiments, a fin ring 548r is formed in the central portion 512 and connected to the baffles 546, where the fin ring 548r is between the shared inflow channel Ip and the bent portions of the baffles 546. In addition, a plurality of fins 548 may be formed in each section to facilities the control in the flow direction of the coolant. For example, five fins 548 are formed in each of the sections, as shown in FIG. 43. However, the number of the fins 548 in each section are not limited thereto, and the number of the fins 548 may be zero, one or more than one. Alternatively, the baffles 546 and/or the fins 548 may extend counterclockwise, the disclosure is not limited thereto.

[0154] Referring to FIG. 44 and FIG. 45, in some embodiments, an inflow channel Ip is formed in the central portion 512 at the center thereof, and one baffle ring 546r is formed on the central portion 512 between the periphery of the central portion 512 and the centered inflow channel Ip to divide the center portion 512 into two independent partition regions (e.g. an inner region and an outer region surrounding the inner region). For example, four baffles 546 are formed on the central portion 512 and connected to the edge of the inflow channel Ip, where the baffles 546 extend from the inflow channel Ip to the baffle ring 546r to divide the inner region into four equal sections. In some embodiments, four outflow channels Op are formed in the four sections of the inner region, respectively. In other words, each section of the inner region has one outflow channel Op and share the same one inflow channel Ip, in some embodiments, as shown in FIG. 44 and FIG. 45. For example, a plurality of fins 548 are formed in each of the sections to facilities the control in the flow direction of the coolant, where the size of the fins 548 are not identical, in part or all. However, the number of the fins 548 in each section are not limited thereto, and the number of the fins 548 may be zero, one or more than one. The size of the fins 548 in the sections of the inner region may be identical. For example, the fins 548 are located between the outflow channel Op and the inflow channel Ip in each section of the inner region, where the fins 548 are parallel to each other and extend in a direction from the additional outflow channel Op to the additional inflow channel Ip (or from the additional inflow channel Ip to the additional outflow channel Op), as shown in FIG. 44 and FIG. 45.

[0155] In some embodiments, additional four inflow channels Ip and additional four outflow channels Op are formed in the outer region of the central portion 512 and arranged in an annular shape side-by-side, and additional four baffles 546 are formed on the central portion 512 and connected to the baffle ring 546r. For example, the additional baffles 546 extend from the baffle ring 546r to the periphery of the central portion 512 (to connect to the flange portion 514), where the additional baffles 546 divide the outer region into four equal sections, and each of the additional baffles 546 is positioned to connect to the edge of one of the additional outflow channels Op. For example, the each of the additional baffles 546 has two separated parts positioned to align with a diameter of one of the additional outflow channels Op, one part is located between the baffle ring 546r and other part is located between the respective one of the additional outflow channels Op and the flange portion 514. In some embodiments, as shown in FIG. 44 and FIG. 45, each section of the outer region has one of the additional inflow channel Ip and shares one of the additional outflow channels Op with another one section immediately adjacent thereto.

[0156] For example, as shown in FIG. 44, a plurality of additional fins 548 are formed between one additional outflow channel Op and one additional inflow channel Ip within the outer region to facilities the control in the flow direction of the coolant, where the additional fins 548 are parallel to each other and extend in a direction from the additional outflow channel Op to the additional inflow channel Ip (or from the additional inflow channel Ip to the additional outflow channel Op).

[0157] However, the disclosure is not limited thereto; in another example, as shown in FIG. 45, a first plurality of additional fins 548 are formed between one additional outflow channel Op and one additional inflow channel Ip within the outer region to facilities the control in the flow direction of the coolant, where the first additional fins 548 are non-parallel to each other and extend in a

direction from the additional outflow channel Op to the additional inflow channel Ip (or from the additional inflow channel Ip to the additional outflow channel Op), and a second plurality of additional fins **548** are formed between the additional inflow channels Ip and the flange portion **514** and extend in a direction passing through the inflow channel Ip located in the inner region. For example, the non-parallel fins **548** have extending axes from ends at a first side thereof are converged while extending axes from other ends at a second side thereof are diverged, where the first side and the second side are opposite along the extending direction of the fins **548**. In FIG. **45**, for instance, the converged ends of the non-parallel fins **548** are closer to the additional inflow channels Ip than the diverged ends, and the diverged ends of the non-parallel fins **548** are closer to the additional outflow channel Op than the converged ends. However, alternatively, the converged ends of the non-parallel fins **548** may be closer to the additional outflow channels Op than the diverged ends, and the diverged ends of the non-parallel fins **548** may be closer to the additional inflow channel Ip than the converged ends.

[0158] The non-limiting arrangements of the microstructures (e.g., **542**, **544**, **546**, or **548**) and the recesses (e.g., R1 through R4) formed on the central portion **512** have been described in detail as shown in FIG. **34** through FIG. **45**. However it is appreciated that besides the uses of the recesses R1 through R4, the arrangements of the microstructures (e.g., **542**, **544**, **546**, or **548**) formed on the central portion **512** may be adopted to the microstructure (e.g., **322**, **324**, **346**, **348**) formed on the metallization layer or a backside portion of the heat dissipating system. Due to the configuration of the heat dissipating module, the heat dissipating system of the disclosure is scalable in correspondence with the size of the semiconductor package bonded therewith.

[0159] In accordance with some embodiments, a package structure includes a wafer-form semiconductor package and a thermal dissipating system. The wafer-form semiconductor package includes semiconductor dies electrically connected with each other. The thermal dissipating system is located on the wafer-form semiconductor package, where the thermal dissipating system has a hollow structure with a fluidic space, and the fluidic space includes a ceiling and a floor. The thermal dissipating system includes at least one inlet opening, at least one outlet opening and a plurality of first microstructures. The at least one inlet opening and the at least one outlet opening are spatially communicated with the fluidic space. The first microstructures are located on the floor, and at least one of the first microstructures is corresponding to the at least one outlet opening.

[0160] In accordance with some embodiments, a package structure includes a wafer-form semiconductor package, a metallization layer, a thermal dissipating module and microstructures. The wafer-form semiconductor package includes dies electrically connected with each other. The metallization layer is located on and thermally coupled to the wafer-form semiconductor package. The thermal dissipating module is located on the metallization layer and includes a plurality of openings, where a confined space is between the thermal dissipating module and the metallization layer and includes one or more fluidic channels providing fluidic communication between the plurality of openings. The microstructures are located on the metallization layer and within the confined space, wherein a first group of the first microstructures is positioned at locations corresponding to the plurality of openings.

[0161] In accordance with some embodiments, a method of manufacturing a package structure includes the following steps, providing a wafer-form semiconductor package comprising semiconductor dies electrically connected with each other; and disposing a heat dissipating system on the wafer-form semiconductor package, comprising: forming a metallization layer over the wafer-form semiconductor package, the metallization layer thermally coupled to the wafer-form semiconductor package; bonding a heat dissipating module having a plurality of openings to the metallization layer to form a confined space between the heat dissipating module and the metallization layer and spatially communicated with the plurality of openings; and disposing first microstructures on the metallization layer within the confined space and at locations corresponding to the plurality of openings, the first microstructure thermally coupled to and bonded to the

metallization layer.

[0162] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the disclosure. Those skilled in the art should appreciate that they may readily use the disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the disclosure.

Claims

1. A package structure, comprising: a semiconductor package, comprising a semiconductor die; and a heat dissipation module, disposed over and thermally coupled to the semiconductor package, wherein the thermal dissipating module with a fluidic space disposed therein, wherein the thermal dissipating module comprises: openings, disposed in and penetrating through a topmost portion of the thermal dissipating module being away from the semiconductor package, wherein in a plane view of the thermal dissipating module, a first group of the openings is disposed at a center of the topmost portion, and a second group of the openings is disposed at an edge of the topmost portion, wherein in the plane view, the fluidic space is entirely divided into a plurality of equal sections, at least one opening of the second group of the openings is respectively disposed in the plurality of equal sections, and the first group of the openings comprises one opening being disposed at a concentric position of and shared by the plurality of equal sections.
2. The package structure of claim 1, wherein in the plane view, each of the plurality of equal sections comprises two or more than two sub-regions being spatially communicated to each other.
3. The package structure of claim 1, wherein in a cross-section view of the heat dissipation module, a first group of recesses are disposed in an inner surface of the topmost portion, and are respectively overlapped with and spatially communicated to the first group of the openings, wherein a thickness of each recess of the first group of recesses is less than a thickness of a respective one opening of the first group of the openings.
4. The package structure of claim 1, wherein in a cross-section view of the heat dissipation module, a first group of recesses are disposed in an inner surface of the topmost portion, and are respectively overlapped with and spatially communicated to the first group of the openings, wherein a lateral size of each recess of the first group of recesses is greater than a lateral size of a respective one opening of the first group of the openings.
5. The package structure of claim 1, wherein in a cross-section view of the heat dissipation module, a second group of recesses are disposed in an inner surface of the topmost portion, and are respectively overlapped with and spatially communicated to the second group of the openings, wherein a thickness of each recess of the second group of recesses is less than a thickness of a respective one opening of the second group of the openings.
6. The package structure of claim 1, wherein in a cross-section view of the heat dissipation module, a second group of recesses are disposed in an inner surface of the topmost portion, and are respectively overlapped with and spatially communicated to the second group of the openings, wherein a lateral size of each recess of the second group of recesses is greater than a lateral size of a respective one opening of the second group of the openings.
7. The package structure of claim 1, wherein in a cross-section view of the heat dissipation module, a third group of recesses are disposed in an inner surface of the topmost portion, and are offset from the first group of the opening and the second group of the openings, wherein a thickness of each recess of the third group of recesses is less than thicknesses of the first group of the openings and the second group of the openings.
8. The package structure of claim 1, wherein in the plane view, a shape of the heat dissipation

module comprises a circle shape, and the plurality of equal sections are in form of sectors.

9. The package structure of claim 1, wherein the semiconductor package further comprises a plurality of conductive terminals electrically coupled to the semiconductor die and an insulating encapsulation encapsulating the semiconductor die, and the plurality of conductive terminals are disposed at a side of the semiconductor package opposite to the heat dissipation module, wherein on a vertical projection in a stacking direction of the semiconductor die and the heat dissipation module, a perimeter of the heat dissipation module is enclosed by a perimeter of the insulating encapsulation comprised in the semiconductor package.

10. The package structure of claim 9, further comprising: a circuit component, electrically coupled to the semiconductor package through the plurality of conductive terminals, wherein the semiconductor package is disposed between the circuit component and the heat dissipation module.

11. The package structure of claim 10, wherein the heat dissipation module is further bonded to the circuit component, and the semiconductor package is disposed in a space confined by the circuit component and the heat dissipating module.

12. A package structure, comprising: a semiconductor package, comprising at least one semiconductor die; and a heat dissipating module, disposed over and thermally coupled to the semiconductor package, wherein the heat dissipating module with a fluidic space disposed therein, wherein the heat dissipating module comprises: openings, disposed in and penetrating through a topmost portion of the heat dissipating module being distant from the semiconductor package, wherein in a plane view of the heat dissipating module, the fluidic space comprises a first fluidic region and a second fluidic region separating from and surrounding the first fluidic region, the first fluidic region is spatially independent from the second fluidic region, and at least one opening of a first group of the openings is disposed at the first fluidic region, and at least one opening of a second group of the openings is disposed at the second fluidic region, wherein a number of the openings in the first fluidic region is greater than or equal to two.

13. The package structure of claim 12, wherein in the plane view, the first fluidic region is in a form of first spiral having a first end and a third end, the second fluidic region is a form of a second spiral having a second end and a fourth end, wherein the first end and the second end are disposed at positions near to a center of the topmost portion, and the third end and the fourth end are disposed at positions near an edge of the topmost portion, wherein the at least one opening of the first group of the openings comprises a first opening of the first group of the openings disposed at the first end, and the at least one opening of the second group of the openings comprises a second opening of the second group of the openings disposed at the second end, wherein a third opening of the first group of the openings is disposed at the fourth end, and a fourth opening of the second group of the openings is disposed at the third end, wherein on a vertical projection in a stacking direction of the at least one semiconductor die and the heat dissipating module, an occupying area of the heat dissipating module is smaller than an occupying area of the semiconductor package.

14. The package structure of claim 12, wherein in the plane view, the at least one opening of the first group of the openings comprises a first opening of the first group of the openings disposed at a center of the first fluidic region, and the at least one opening of the second group of the openings comprises second openings of the second group of the openings disposed at the second fluidic region and distributed along an edge thereof, wherein third openings of the first group of the openings are disposed at the second fluidic region and distributed along an edge of the first fluidic region, and fourth openings of the second group of the openings are disposed at the first fluidic region and distributed along the edge of the first fluidic region.

15. The package structure of claim 12, wherein the first fluidic region comprises a plurality of first equal sections, and the second fluidic region comprises a plurality of second equal section, wherein in the plane view, a shape of the heat dissipating module comprises a circle shape, and the plurality of first equal sections and the plurality of second equal sections are in form of sectors.

16. The package structure of claim 15, wherein in the plane view, each of the plurality of first equal

sections comprises two or more than two sub-regions being spatially communicated to each other, and each of the plurality of second equal sections comprises two or more than two sub-regions being spatially communicated to each other.

17. The package structure of claim 15, wherein in the plane view, the at least one opening of the first group of the openings comprises a first opening of the first group of the openings disposed at a center of the first fluidic region, and the at least one opening of the second group of the openings comprises second openings of the second group of the openings disposed at the second fluidic region and distributed along an edge thereof, wherein third openings of the first group of the openings are disposed at the second fluidic region and distributed along the edge thereof, and are alternatively arranged with the second openings, wherein fourth openings of the second group of the openings are disposed at the first fluidic region and distributed along an edge thereof.

18. A package structure, comprising: a semiconductor package, comprising a semiconductor die; and a thermal dissipating system, disposed on and thermally coupled to the semiconductor package, the thermal dissipating system being in direct physical contact with the semiconductor die, wherein the thermal dissipating system with a fluidic space disposed herein, wherein the thermal dissipating system comprises at least one inlet opening and at least one outlet opening spatially communicated with the fluidic space.

19. The package structure of claim 18, wherein the thermal dissipating system further comprises: a conductive layer, disposed on and thermally coupled to the semiconductor package; and a heat dissipating module, comprising a lid having a central portion and a flange portion bonded to a periphery of the central portion, wherein the flange portion is bonded to the conductive layer to confine the fluidic space between the conductive layer, the flange portion and the central portion.

20. The package structure of claim 18, wherein the thermal dissipating system further comprises: a conductive layer, disposed on and thermally coupled to the semiconductor package; a plurality of conductive connectors; and a heat dissipating module, bonded to and thermally coupled to the conductive layer through the plurality of the conductive connectors, and comprising a block having a top-side portion, a bottom-side portion and a side portion connecting the top-side portion and the bottom-side portion, wherein the fluidic space is confined by the bottom-side portion, the side portion and the top-side portion.
